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JEE MAIN · MATHEMATICS · CHAPTER-WISE PYQS

Differential Equations

JEE Main Mathematics · Previous Year Questions with Answer Key

266	196	2002 – 2026	23	English
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MCQ (Single Correct Answer) 193 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

Let $y = y(x)$ be the solution of the differential equation $x\sqrt{1-x^2}dy + \left(y\sqrt{1-x^2} - x\cos^{-1}x\right)dx = 0, x \in (0,1), \lim_{x\to 1^-} y(x) = 1$. Then $y\left(\frac{1}{2}\right)$ equals :

(A) $3 - \frac{\pi}{\sqrt{3}}$ (B) $4 - \sqrt{3}\pi$ (C) $4 - \frac{2\pi}{\sqrt{3}}$ (D) $3 - \frac{\pi}{2\sqrt{3}}$

Open solution

Q2. JEE Main 2026 (Online) 6th April Evening Shift

Let $f : \mathbf{R} \to \mathbf{R}$ be such that $f(xy) = f(x)f(y)$, for all $x, y \in \mathbf{R}$ and $f(0) \neq 0$. Let $g : [1,\infty) \to \mathbf{R}$ be a differentiable function such that

$x^2 g(x) = \int_1^x \left(t^2 f(t) - tg(t)\right)dt$

Then $g(2)$ is equal to :

(A) $\frac{13}{8}$ (B) $\frac{11}{16}$ (C) $\frac{15}{32}$ (D) $\frac{17}{64}$

Open solution

Q3. JEE Main 2026 (Online) 5th April Evening Shift

Let $f : [1,\infty) \to \mathbf{R}$ be a differentiable function defined as $f(x) = \int_1^x f(t)dt + (1-x)\left(\log_e x - 1\right) + e$.

Then the value of $f(f(1))$ is :

(A) $(1 + e^e)$ (B) $(1 + e)$

(C) $(1 + e + e^e)$ (D) $1 + 2e$

Open solution

Q4. JEE Main 2026 (Online) 4th April Evening Shift

Let $y = y(x)$ be the solution of the differential equation :

$\frac{dy}{dx} + \left(\frac{6x^2 + (3x^2 + 2x^3 + 4)e^{-2x}}{(x^3+2)(2+e^{-2x})}\right)y = 2 + e^{-2x}$

$x \in (-1,2)$, satisfying $y(0) = \frac{3}{2}$. If $y(1) = \alpha\left(2 + e^{-2}\right)$, then α is equal to :

(A) $\frac{13}{8}$ (B) $\frac{6}{13}$ (C) $\frac{12}{13}$ (D) $\frac{13}{12}$

Open solution

Q5. JEE Main 2026 (Online) 4th April Morning Shift

Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} = \left(1 + x + x^2\right)\left(1 - y + y^2\right), y(0) = \frac{1}{2}$.
Then $(2y(1) - 1)$ is equal to

(A) $\sqrt{3}\tan\left(\frac{11\sqrt{3}}{6}\right)$ (B) $\frac{\sqrt{3}}{2}\tan\left(\frac{11\sqrt{3}}{12}\right)$

(C) $\sqrt{3}\tan\left(\frac{11\sqrt{3}}{12}\right)$ (D) $\frac{\sqrt{3}}{2}\tan\left(\frac{11\sqrt{3}}{6}\right)$

Open solution

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Q6. JEE Main 2026 (Online) 2nd April Evening Shift

Let $x = x(y)$ be the solution of the differential equation $2y^2\frac{dx}{dy} - 2xy + x^2 = 0, y > 1, x(e) = e$.

Then $x(e^2)$ is equal to:

(A) $\frac{3}{2}e^2$ (B) $\frac{2}{3}e^2$ (C) e^2 (D) $2e^2$

Open solution

Q7. JEE Main 2026 (Online) 2nd April Morning Shift

If the curve $y = f(x)$ passes through the point $(1, e)$ and satisfies the differential equation $dy = y(2 + \log_e x)dx, x > 0$, then $f(e)$ is equal to :

(A) e^e (B) e^{e^2} (C) e^{2e} (D) e^{2^e}

Open solution

Q8. JEE Main 2026 (Online) 2nd April Morning Shift

Let $y = y(x)$ be the solution curve of the differential equation

$(1 + \sin x)\dfrac{dy}{dx} + (y + 1)\cos x = 0,\ y(0) = 0$. If the curve $y = y(x)$ passes through the point $\left(\alpha, \dfrac{-1}{2}\right)$,

then a value of α is:

(A) $\dfrac{\pi}{6}$ (B) $\dfrac{\pi}{4}$ (C) $\dfrac{\pi}{3}$ (D) $\dfrac{\pi}{2}$

Open solution

Q9. JEE Main 2026 (Online) 28th January Evening Shift

Let $y = y(x)$ be the solution of the differential equation $x\frac{dy}{dx} - y = x^2\cot x, x \in (0, \pi)$. If $y\left(\frac{\pi}{2}\right) = \frac{\pi}{2}$, then

$6y\left(\frac{\pi}{6}\right) - 8y\left(\frac{\pi}{4}\right)$ is equal to :

(A) -3π (B) 3π (C) $-\pi$ (D) π

Open solution

Q10. JEE Main 2026 (Online) 28th January Morning Shift

Let $y = y(x)$ be the solution of the differential equation

$x\frac{dy}{dx} - \sin 2y = x^3\left(2 - x^3\right)\cos^2 y, x \neq 0$.

If $y(2) = 0$, then $\tan(y(1))$ is equal to

(A) $-\frac{7}{4}$ (B) $-\frac{3}{4}$ (C) $\frac{3}{4}$ (D) $\frac{7}{4}$

Open solution

Q11. JEE Main 2026 (Online) 23rd January Morning Shift

Let $y = y(x)$ be the solution of the differential equation

$x^4\,dy + \left(4x^3y + 2\sin x\right)dx = 0, x > 0, y\left(\frac{\pi}{2}\right) = 0$.

Then $\pi^4 y\left(\frac{\pi}{3}\right)$ is equal to :

(A) 92 (B) 72 (C) 64 (D) 81

Open solution

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Q12. JEE Main 2026 (Online) 22nd January Evening Shift

If $y = y(x)$ satisfies the differential equation $16(\sqrt{x+9\sqrt{x}})(4+\sqrt{9+\sqrt{x}})\cos y\,dy = (1+2\sin y)dx, x > 0$ and $y(256) = \frac{\pi}{2}, y(49) = \alpha$, then $2\sin\alpha$ is equal to :

Open solution

(A) $2\sqrt{2}-1$ (B) $\sqrt{2}-1$ (C) $2(\sqrt{2}-1)$ (D) $3(\sqrt{2}-1)$

Q13. JEE Main 2026 (Online) 22nd January Morning Shift

Let the solution curve of the differential equation $xdy - ydx = \sqrt{x^2+y^2}dx, x > 0, y(1) = 0$, be $y = y(x)$. Then $y(3)$ is equal to

Open solution

(A) 6 (B) 4 (C) 1 (D) 2

Q14. JEE Main 2026 (Online) 21st January Evening Shift

Let $y = y(x)$ be the solution of the differential equation $\sec x\frac{dy}{dx} - 2y = 2 + 3\sin x, x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, $y(0) = -\frac{7}{4}$. Then $y\left(\frac{\pi}{6}\right)$ is equal to :

Open solution

(A) $-\frac{5}{2}$ (B) $-3\sqrt{2}-7$ (C) $-\frac{5}{4}$ (D) $-3\sqrt{3}-7$

Q15. JEE Main 2026 (Online) 21st January Morning Shift

Let $y = y(x)$ be the solution curve of the differential equation $(1+x^2)dy + (y - \tan^{-1}x)dx = 0, y(0) = 1$. Then the value of $y(1)$ is :

Open solution

(A) $\frac{4}{e^{\pi/4}} - \frac{\pi}{2} - 1$ (B) $\frac{2}{e^{\pi/4}} + \frac{\pi}{4} - 1$
(C) $\frac{4}{e^{\pi/4}} + \frac{\pi}{2} - 1$ (D) $\frac{2}{e^{\pi/4}} - \frac{\pi}{4} - 1$

Q16. JEE Main 2025 (Online) 8th April Evening Shift

Let $f(x) = x - 1$ and $g(x) = e^x$ for $x \in \mathbb{R}$. If $\frac{dy}{dx} = \left(e^{-2\sqrt{x}}g(f(f(x))) - \frac{y}{\sqrt{x}}\right), y(0) = 0$, then $y(1)$ is

Open solution

(A) $\frac{1-e^3}{e^4}$ (B) $\frac{e-1}{e^4}$ (C) $\frac{1-e^2}{e^4}$ (D) $\frac{2e-1}{e^3}$

Q17. JEE Main 2025 (Online) 7th April Evening Shift

Let y = y(x) be the solution of the differential equation $(x^2+1)y' - 2xy = (x^4+2x^2+1)\cos x$, $y(0) = 1$. Then $\int_{-3}^{3} y(x)\,dx$ is :

Open solution

(A) 36 (B) 24 (C) 18 (D) 30

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Q18. JEE Main 2025 (Online) 7th April Morning Shift

Let $y = y(x)$ be the solution curve of the differential equation

$x\left(x^2 + e^x\right)dy + \left(e^x(x-2)y - x^3\right)dx = 0, x > 0$, passing through the point $(1, 0)$. Then $y(2)$ is equal to :

Open solution

(A) $\frac{2}{2+e^2}$ (B) $\frac{4}{4-e^2}$ (C) $\frac{4}{4+e^2}$ (D) $\frac{2}{2-e^2}$

Q19. JEE Main 2025 (Online) 4th April Evening Shift

If a curve $y = y(x)$ passes through the point $\left(1, \frac{\pi}{2}\right)$ and satisfies the differential equation $\left(7x^4 \cot y - e^x \operatorname{cosec} y\right)\frac{dx}{dy} = x^5, x \geq 1$, then at $x = 2$, the value of $\cos y$ is :

Open solution

(A) $\frac{2e^2+e}{64}$ (B) $\frac{2e^2-e}{64}$
(C) $\frac{2e^2-e}{128}$ (D) $\frac{2e^2+e}{128}$

Q20. JEE Main 2025 (Online) 3rd April Evening Shift

Let $y = y(x)$ be the solution of the differential equation

$\frac{dy}{dx} + 3\left(\tan^2 x\right)y + 3y = \sec^2 x, y(0) = \frac{1}{3} + e^3$. Then $y\left(\frac{\pi}{4}\right)$ is equal to :

Open solution

(A) $\frac{4}{3}$ (B) $\frac{2}{3} + e^3$ (C) $\frac{4}{3} + e^3$ (D) $\frac{2}{3}$

Q21. JEE Main 2025 (Online) 3rd April Morning Shift

Let g be a differentiable function such that $\int_0^x g(t)dt = x - \int_0^x tg(t)dt, x \geq 0$ and let $y = y(x)$ satisfy the differential equation $\frac{dy}{dx} - y\tan x = 2(x+1)\sec x g(x), x \in \left[0, \frac{\pi}{2}\right)$. If $y(0) = 0$, then $y\left(\frac{\pi}{3}\right)$ is equal to

Open solution

(A) $\frac{4\pi}{3}$ (B) $\frac{2\pi}{3}$ (C) $\frac{2\pi}{3\sqrt{3}}$ (D) $\frac{4\pi}{3\sqrt{3}}$

Q22. JEE Main 2025 (Online) 29th January Evening Shift

If for the solution curve $y = f(x)$ of the differential equation $\frac{dy}{dx} + (\tan x)y = \frac{2+\sec x}{(1+2\sec x)^2}$, $x \in \left(\frac{-\pi}{2}, \frac{\pi}{2}\right), f\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{10}$, then $f\left(\frac{\pi}{4}\right)$ is equal to:

Open solution

(A) $\frac{5-\sqrt{3}}{2\sqrt{2}}$ (B) $\frac{4-\sqrt{2}}{14}$
(C) $\frac{9\sqrt{3}+3}{10(4+\sqrt{3})}$ (D) $\frac{\sqrt{3}+1}{10(4+\sqrt{3})}$

Q23. JEE Main 2025 (Online) 29th January Morning Shift

Let *y = y(x)* be the solution of the differential equation :

$\cos x(\log_e(\cos x))^2 dy + (\sin x - 3y\sin x\log_e(\cos x))dx = 0$, x ∈ (0, $\frac{\pi}{2}$). If $y(\frac{\pi}{4}) = -\frac{1}{\log_e 2}$, then $y(\frac{\pi}{6})$ is equal to :

Open solution

(A) $\frac{2}{\log_e(3)-\log_e(4)}$ (B) $-\frac{1}{\log_e(4)}$
(C) $\frac{1}{\log_e(4)-\log_e(3)}$ (D) $\frac{1}{\log_e(3)-\log_e(4)}$

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Q24. JEE Main 2025 (Online) 28th January Morning Shift

Let for some function $y = f(x)$, $\int_0^x tf(t)dt = x^2 f(x), x > 0$ and $f(2) = 3$. Then $f(6)$ is equal to

(A) 1 (B) 6 (C) 2 (D) 3

Open solution

Q25. JEE Main 2025 (Online) 24th January Morning Shift

Let $y = y(x)$ be the solution of the differential equation $\left(xy - 5x^2\sqrt{1+x^2}\right)dx + \left(1+x^2\right)dy = 0, y(0) = 0$. Then $y(\sqrt{3})$ is equal to

(A) $\frac{5\sqrt{3}}{2}$ (B) $\sqrt{\frac{15}{2}}$ (C) $\sqrt{\frac{14}{3}}$ (D) $2\sqrt{2}$

Open solution

Q26. JEE Main 2025 (Online) 23rd January Evening Shift

Let $x = x(y)$ be the solution of the differential equation $y = \left(x - y\frac{dx}{dy}\right)\sin\left(\frac{x}{y}\right), y > 0$ and $x(1) = \frac{\pi}{2}$. Then $\cos(x(2))$ is equal to :

(A) $2\left(\log_e 2\right) - 1$ (B) $1 - 2(\log_e 2)^2$

(C) $1 - 2\left(\log_e 2\right)$ (D) $2(\log_e 2)^2 - 1$

Open solution

Q27. JEE Main 2025 (Online) 23rd January Morning Shift

Let a curve $y = f(x)$ pass through the points $(0, 5)$ and $(\log_e 2, k)$. If the curve satisfies the differential equation $2(3+y)e^{2x}dx - \left(7+e^{2x}\right)dy = 0$, then k is equal to

(A) 32 (B) 8 (C) 4 (D) 16

Open solution

Q28. JEE Main 2025 (Online) 22nd January Evening Shift

If $x = f(y)$ is the solution of the differential equation $\left(1+y^2\right) + \left(x - 2e^{\tan^{-1}y}\right)\frac{dy}{dx} = 0, y \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ with $f(0) = 1$, then $f\left(\frac{1}{\sqrt{3}}\right)$ is equal to :

(A) $e^{\pi/4}$ (B) $e^{\pi/12}$ (C) $e^{\pi/6}$ (D) $e^{\pi/3}$

Open solution

Q29. JEE Main 2025 (Online) 22nd January Morning Shift

Let $x = x(y)$ be the solution of the differential equation $y^2\,dx + \left(x - \frac{1}{y}\right)dy = 0$. If $x(1) = 1$, then $x\left(\frac{1}{2}\right)$ is :

(A) $\frac{3}{2} + e$ (B) $\frac{1}{2} + e$ (C) $3 + e$ (D) $3 - e$

Open solution

Q30. JEE Main 2025 (Online) 22nd January Morning Shift

Let $f(x)$ be a real differentiable function such that $f(0) = 1$ and $f(x+y) = f(x)f'(y) + f'(x)f(y)$ for all $x, y \in \mathbf{R}$. Then $\sum_{n=1}^{100} \log_e f(n)$ is equal to :

(A) 2406 (B) 5220 (C) 2525 (D) 2384

Open solution

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Q31. JEE Main 2025 (Online) 22nd January Morning Shift

Let $f : \mathbf{R} \to \mathbf{R}$ be a twice differentiable function such that $f(x+y) = f(x)f(y)$ for all $x, y \in \mathbf{R}$. If $f'(0) = 4\mathrm{a}$ and f satisfies $f''(x) - 3\mathrm{a}f'(x) - f(x) = 0, \mathrm{a} > 0$, then the area of the region $\mathrm{R} = \{(x, y) \mid 0 \le y \le f(ax), 0 \le x \le 2\}$ is :

(A) $\mathrm{e}^2 - 1$ (B) $e^4 + 1$ (C) $\mathrm{e}^2 + 1$ (D) $e^4 - 1$

Open solution

Q32. JEE Main 2024 (Online) 9th April Evening Shift

Let $\int_0^x \sqrt{1 - (y'(t))^2}\,dt = \int_0^x y(t)dt, 0 \le x \le 3, y \ge 0, y(0) = 0$. Then at $x = 2, y'' + y + 1$ is equal to

(A) $\sqrt{2}$ (B) 2 (C) 1/2 (D) 1

Open solution

Q33. JEE Main 2024 (Online) 9th April Morning Shift

The solution of the differential equation $(x^2 + y^2)\mathrm{d}x - 5xy\,\mathrm{d}y = 0, y(1) = 0$, is :

(A) $\left|x^2 - 4y^2\right|^5 = x^2$ (B) $\left|x^2 - 2y^2\right|^6 = x$ (C) $\left|x^2 - 2y^2\right|^5 = x^2$ (D) $\left|x^2 - 4y^2\right|^6 = x$

Open solution

Q34. JEE Main 2024 (Online) 9th April Morning Shift

The solution curve, of the differential equation $2y\frac{\mathrm{d}y}{\mathrm{d}x} + 3 = 5\frac{\mathrm{d}y}{\mathrm{d}x}$, passing through the point $(0, 1)$ is a conic, whose vertex lies on the line :

(A) $2x + 3y = -9$ (B) $2x + 3y = -6$ (C) $2x + 3y = 9$ (D) $2x + 3y = 6$

Open solution

Q35. JEE Main 2024 (Online) 8th April Evening Shift

Let $y = y(x)$ be the solution curve of the differential equation $\sec y\frac{\mathrm{d}y}{\mathrm{d}x} + 2x\sin y = x^3\cos y, y(1) = 0$. Then $y(\sqrt{3})$ is equal to:

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{12}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{4}$

Open solution

Q36. JEE Main 2024 (Online) 8th April Morning Shift

Let $f(x)$ be a positive function such that the area bounded by $y = f(x), y = 0$ from $x = 0$ to $x = a > 0$ is $e^{-a} + 4a^2 + a - 1$. Then the differential equation, whose general solution is $y = c_1 f(x) + c_2$, where c_1 and c_2 are arbitrary constants, is

(A) $(8e^x + 1)\frac{d^2y}{dx^2} - \frac{dy}{dx} = 0$ (B) $(8e^x + 1)\frac{d^2y}{dx^2} + \frac{dy}{dx} = 0$
(C) $(8e^x - 1)\frac{d^2y}{dx^2} - \frac{dy}{dx} = 0$ (D) $(8e^x - 1)\frac{d^2y}{dx^2} + \frac{dy}{dx} = 0$

Open solution

Q37. JEE Main 2024 (Online) 8th April Morning Shift

Let $y = y(x)$ be the solution of the differential equation $(1 + y^2)e^{\tan x}dx + \cos^2 x(1 + e^{2\tan x})dy = 0, y(0) = 1$. Then $y\left(\frac{\pi}{4}\right)$ is equal to

(A) $\frac{1}{e^2}$ (B) $\frac{2}{e^2}$ (C) $\frac{2}{e}$ (D) $\frac{1}{e}$

Open solution

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Q38. JEE Main 2024 (Online) 6th April Evening Shift

Suppose the solution of the differential equation $\frac{dy}{dx} = \frac{(2+\alpha)x-\beta y+2}{\beta x-2\alpha y-(\beta\gamma-4\alpha)}$ represents a circle passing through origin. Then the radius of this circle is :

(A) $\sqrt{17}$ (B) 2 (C) $\frac{\sqrt{17}}{2}$ (D) $\frac{1}{2}$

Open solution

Q39. JEE Main 2024 (Online) 6th April Morning Shift

Let $y = y(x)$ be the solution of the differential equation $(2x\log_e x)\frac{dy}{dx} + 2y = \frac{3}{x}\log_e x, x > 0$ and $y\left(e^{-1}\right) = 0$. Then, $y(e)$ is equal to

(A) $-\frac{3}{e}$ (B) $-\frac{3}{2e}$ (C) $-\frac{2}{3e}$ (D) $-\frac{2}{e}$

Open solution

Q40. JEE Main 2024 (Online) 6th April Morning Shift

Let $y = y(x)$ be the solution of the differential equation $\left(1+x^2\right)\frac{dy}{dx} + y = e^{\tan^{-1}x}$, $y(1) = 0$. Then $y(0)$ is

(A) $\frac{1}{4}\left(e^{\pi/2}-1\right)$ (B) $\frac{1}{2}\left(1-e^{\pi/2}\right)$
(C) $\frac{1}{4}\left(1-e^{\pi/2}\right)$ (D) $\frac{1}{2}\left(e^{\pi/2}-1\right)$

Open solution

Q41. JEE Main 2024 (Online) 5th April Evening Shift

The differential equation of the family of circles passing through the origin and having centre at the line $y = x$ is :

(A) $\left(x^2-y^2+2xy\right)dx = \left(x^2-y^2+2xy\right)dy$ (B) $\left(x^2+y^2-2xy\right)dx = \left(x^2+y^2+2xy\right)dy$
(C) $\left(x^2+y^2+2xy\right)dx = \left(x^2+y^2-2xy\right)dy$ (D) $\left(x^2-y^2+2xy\right)dx = \left(x^2-y^2-2xy\right)dy$

Open solution

Q42. JEE Main 2024 (Online) 5th April Morning Shift

If $y = y(x)$ is the solution of the differential equation $\frac{dy}{dx} + 2y = \sin(2x), y(0) = \frac{3}{4}$, then $y\left(\frac{\pi}{8}\right)$ is equal to :

(A) $e^{-\pi/8}$ (B) $e^{\pi/4}$ (C) $e^{-\pi/4}$ (D) $e^{\pi/8}$

Open solution

Q43. JEE Main 2024 (Online) 4th April Evening Shift

Let $y = y(x)$ be the solution of the differential equation $(x^2+4)^2dy + (2x^3y+8xy-2)dx = 0$. If $y(0) = 0$, then $y(2)$ is equal to

(A) 2π (B) $\frac{\pi}{8}$ (C) $\frac{\pi}{16}$ (D) $\frac{\pi}{32}$

Open solution

Q44. JEE Main 2024 (Online) 4th April Morning Shift

If the solution $y = y(x)$ of the differential equation $(x^4+2x^3+3x^2+2x+2)dy - (2x^2+2x+3)dx = 0$ satisfies $y(-1) = -\frac{\pi}{4}$, then $y(0)$ is equal to :

(A) $-\frac{\pi}{12}$ (B) $\frac{\pi}{2}$ (C) 0 (D) $\frac{\pi}{4}$

Open solution

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Q45. JEE Main 2024 (Online) 1st February Evening Shift

Let α be a non-zero real number. Suppose $f : \mathbf{R} \to \mathbf{R}$ is a differentiable function such that $f(0) = 2$ and $\lim\limits_{x \to -\infty} f(x) = 1$. If $f'(x) = \alpha f(x) + 3$, for all $x \in \mathbf{R}$, then $f(-\log_e 2)$ is equal to :

(A) 7 (B) 9 (C) 3 (D) 5

Open solution

Q46. JEE Main 2024 (Online) 1st February Morning Shift

Let $y = y(x)$ be the solution of the differential equation

$\frac{dy}{dx} = 2x(x+y)^3 - x(x+y) - 1, y(0) = 1.$

Then, $\left(\frac{1}{\sqrt{2}} + y\left(\frac{1}{\sqrt{2}}\right)\right)^2$ equals :

(A) $\frac{4}{4+\sqrt{e}}$ (B) $\frac{3}{3-\sqrt{e}}$ (C) $\frac{2}{1+\sqrt{e}}$ (D) $\frac{1}{2-\sqrt{e}}$

Open solution

Q47. JEE Main 2024 (Online) 31st January Evening Shift

The temperature $T(t)$ of a body at time $t = 0$ is 160°F and it decreases continuously as per the differential equation $\frac{dT}{dt} = -K(T - 80)$, where K is a positive constant. If $T(15) = 120^\circ$F, then $T(45)$ is equal to

(A) 90° F (B) 85° F (C) 80° F (D) 95° F

Open solution

Q48. JEE Main 2024 (Online) 31st January Morning Shift

Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} = \frac{(\tan x) + y}{\sin x(\sec x - \sin x \tan x)}, x \in \left(0, \frac{\pi}{2}\right)$ satisfying the condition $y\left(\frac{\pi}{4}\right) = 2$. Then, $y\left(\frac{\pi}{3}\right)$ is

(A) $\sqrt{3}(2 + \log_e 3)$ (B) $\sqrt{3}(1 + 2\log_e 3)$

(C) $\sqrt{3}\left(2 + \log_e \sqrt{3}\right)$ (D) $\frac{\sqrt{3}}{2}(2 + \log_e 3)$

Open solution

Q49. JEE Main 2024 (Online) 31st January Morning Shift

The solution curve of the differential equation $y\frac{dx}{dy} = x(\log_e x - \log_e y + 1), x > 0, y > 0$ passing through the point $(e, 1)$ is

(A) $\left|\log_e \frac{y}{x}\right| = y^2$ (B) $\left|\log_e \frac{y}{x}\right| = x$

(C) $\left|\log_e \frac{x}{y}\right| = y$ (D) $2\left|\log_e \frac{x}{y}\right| = y + 1$

Open solution

Q50. JEE Main 2024 (Online) 30th January Morning Shift

Let $y = y(x)$ be the solution of the differential equation $\sec x\, dy + \{2(1-x)\tan x + x(2-x)\}dx = 0$ such that $y(0) = 2$. Then $y(2)$ is equal to:

(A) $2\{\sin(2) + 1\}$ (B) 2 (C) 1 (D) $2\{1 - \sin(2)\}$

Open solution

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Q51. JEE Main 2024 (Online) 29th January Evening Shift

If $\sin\left(\frac{y}{x}\right) = \log_e|x| + \frac{\alpha}{2}$ is the solution of the differential equation $x\cos\left(\frac{y}{x}\right)\frac{dy}{dx} = y\cos\left(\frac{y}{x}\right) + x$ and $y(1) = \frac{\pi}{3}$, then α^2 is equal to

(A) 12 (B) 9 (C) 4 (D) 3

Open solution

Q52. JEE Main 2024 (Online) 29th January Morning Shift

A function $y = f(x)$ satisfies $f(x)\sin 2x + \sin x - \left(1 + \cos^2 x\right)f'(x) = 0$ with condition $f(0) = 0$. Then, $f\left(\frac{\pi}{2}\right)$ is equal to

(A) 2 (B) 1 (C) -1 (D) 0

Open solution

Q53. JEE Main 2024 (Online) 27th January Evening Shift

If $y = y(x)$ is the solution curve of the differential equation $\left(x^2 - 4\right)\mathrm{d}y - \left(y^2 - 3y\right)\mathrm{d}x = 0, x > 2, y(4) = \frac{3}{2}$ and the slope of the curve is never zero, then the value of $y(10)$ equals :

(A) $\frac{3}{1+(8)^{1/4}}$ (B) $\frac{3}{1-(8)^{1/4}}$ (C) $\frac{3}{1-2\sqrt{2}}$ (D) $\frac{3}{1+2\sqrt{2}}$

Open solution

Q54. JEE Main 2024 (Online) 27th January Morning Shift

Let $x = x(\mathrm{t})$ and $y = y(\mathrm{t})$ be solutions of the differential equations $\frac{\mathrm{dx}}{\mathrm{dt}} + \mathrm{ax} = 0$ and $\frac{\mathrm{dy}}{\mathrm{dt}} + \mathrm{by} = 0$ respectively, $\mathrm{a}, \mathrm{b} \in \mathbf{R}$. Given that $x(0) = 2; y(0) = 1$ and $3y(1) = 2x(1)$, the value of t, for which $x(\mathrm{t}) = y(\mathrm{t})$, is :

(A) $\log_{\frac{2}{3}} 2$ (B) $\log_{\frac{4}{3}} 2$ (C) $\log_4 3$ (D) $\log_3 4$

Open solution

Q55. JEE Main 2023 (Online) 15th April Morning Shift

Let $x = x(y)$ be the solution of the differential equation

$2(y+2)\log_e(y+2)dx + (x + 4 - 2\log_e(y+2))dy = 0, y > -1$

with $x\left(e^4 - 2\right) = 1$. Then $x\left(e^9 - 2\right)$ is equal to :

(A) $\frac{4}{9}$ (B) $\frac{32}{9}$ (C) $\frac{10}{3}$ (D) 3

Open solution

Q56. JEE Main 2023 (Online) 13th April Morning Shift

Let $y = y_1(x)$ and $y = y_2(x)$ be the solution curves of the differential equation $\frac{dy}{dx} = y + 7$ with initial conditions $y_1(0) = 0$ and $y_2(0) = 1$ respectively. Then the curves $y = y_1(x)$ and $y = y_2(x)$ intersect at

(A) no point (B) two points

(C) infinite number of points (D) one point

Open solution

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Q57. JEE Main 2023 (Online) 12th April Morning Shift

Let $y = y(x), y > 0$, be a solution curve of the differential equation $(1+x^2)\mathrm{d}y = y(x-y)\mathrm{d}x$. If $y(0) = 1$ and $y(2\sqrt{2}) = \beta$, then

(A) $e^{\beta^{-1}} = e^{-2}(3+2\sqrt{2})$

(B) $e^{3\beta^{-1}} = e(5+\sqrt{2})$

(C) $e^{3\beta^{-1}} = e(3+2\sqrt{2})$

(D) $e^{\beta^{-1}} = e^{-2}(5+\sqrt{2})$

Open solution

Q58. JEE Main 2023 (Online) 11th April Evening Shift

Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} + \frac{5}{x(x^5+1)}y = \frac{(x^5+1)^2}{x^7}, x > 0$. If $y(1) = 2$, then $y(2)$ is equal to :

(A) $\frac{693}{128}$

(B) $\frac{697}{128}$

(C) $\frac{637}{128}$

(D) $\frac{679}{128}$

Open solution

Q59. JEE Main 2023 (Online) 11th April Morning Shift

Let $y = y(x)$ be a solution curve of the differential equation

$(1-x^2y^2)dx = ydx + xdy$.

If the line $x = 1$ intersects the curve $y = y(x)$ at $y = 2$ and the line $x = 2$ intersects the curve $y = y(x)$ at $y = \alpha$, then a value of α is :

(A) $\frac{1+3e^2}{2(3e^2-1)}$

(B) $\frac{3e^2}{2(3e^2-1)}$

(C) $\frac{1-3e^2}{2(3e^2+1)}$

(D) $\frac{3e^2}{2(3e^2+1)}$

Open solution

Q60. JEE Main 2023 (Online) 10th April Morning Shift

Let f be a differentiable function such that $x^2f(x) - x = 4\int_0^x tf(t)dt$, $f(1) = \frac{2}{3}$. Then $18f(3)$ is equal to :

(A) 160

(B) 210

(C) 150

(D) 180

Open solution

Q61. JEE Main 2023 (Online) 6th April Evening Shift

If the solution curve $f(x, y) = 0$ of the differential equation

$(1+\log_e x)\frac{dx}{dy} - x\log_e x = e^y, x > 0$,

passes through the points $(1, 0)$ and $(\alpha, 2)$, then α^α is equal to :

(A) $e^{\sqrt{2}e^2}$

(B) $e^{2e^{\sqrt{2}}}$

(C) e^{e^2}

(D) e^{2e^2}

Open solution

Q62. JEE Main 2023 (Online) 1st February Evening Shift

Let $\alpha x = \exp\left(x^\beta y^\gamma\right)$ be the solution of the differential equation $2x^2y\,\mathrm{d}y - \left(1-xy^2\right)\mathrm{d}x = 0, x > 0, y(2) = \sqrt{\log_e 2}$. Then $\alpha+\beta-\gamma$ equals :

(A) 1

(B) 0

(C) 3

(D) -1

Open solution

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Q63. JEE Main 2023 (Online) 1st February Morning Shift

Open solution

The area enclosed by the closed curve C given by the differential equation

$\frac{dy}{dx}+\frac{x+a}{y-2}=0, y(1)=0$ is 4π.

Let P and Q be the points of intersection of the curve C and the y-axis. If normals at P and Q on the curve C intersect x-axis at points R and S respectively, then the length of the line segment RS is :

(A) $\frac{4\sqrt{3}}{3}$ (B) $2\sqrt{3}$ (C) 2 (D) $\frac{2\sqrt{3}}{3}$

Q64. JEE Main 2023 (Online) 1st February Morning Shift

Open solution

If $y=y(x)$ is the solution curve of the differential equation

$\frac{dy}{dx}+y\tan x=x\sec x, 0\le x\le\frac{\pi}{3}, y(0)=1$, then $y\left(\frac{\pi}{6}\right)$ is equal to

(A) $\frac{\pi}{12}-\frac{\sqrt{3}}{2}\log_e\left(\frac{2\sqrt{3}}{e}\right)$ (B) $\frac{\pi}{12}+\frac{\sqrt{3}}{2}\log_e\left(\frac{2\sqrt{3}}{e}\right)$

(C) $\frac{\pi}{12}+\frac{\sqrt{3}}{2}\log_e\left(\frac{2}{e\sqrt{3}}\right)$ (D) $\frac{\pi}{12}-\frac{\sqrt{3}}{2}\log_e\left(\frac{2}{e\sqrt{3}}\right)$

Q65. JEE Main 2023 (Online) 31st January Evening Shift

Open solution

Let $y=y(x)$ be the solution of the differential equation

$\left(3y^2-5x^2\right)y\,dx+2x\left(x^2-y^2\right)dy=0$

such that $y(1)=1$. Then $\left|(y(2))^3-12y(2)\right|$ is equal to :

(A) 64 (B) $16\sqrt{2}$ (C) 32 (D) $32\sqrt{2}$

Q66. JEE Main 2023 (Online) 30th January Evening Shift

Open solution

The solution of the differential equation

$\frac{dy}{dx}=-\left(\frac{x^2+3y^2}{3x^2+y^2}\right), y(1)=0$ is :

(A) $\log_e|x+y|+\frac{xy}{(x+y)^2}=0$ (B) $\log_e|x+y|-\frac{xy}{(x+y)^2}=0$

(C) $\log_e|x+y|+\frac{2xy}{(x+y)^2}=0$ (D) $\log_e|x+y|-\frac{2xy}{(x+y)^2}=0$

Q67. JEE Main 2023 (Online) 30th January Morning Shift

Open solution

Let the solution curve $y=y(x)$ of the differential equation

$\frac{dy}{dx}-\frac{3x^5\tan^{-1}(x^3)}{(1+x^6)^{3/2}}y=2x\exp\left\{\frac{x^3-\tan^{-1}x^3}{\sqrt{(1+x^6)}}\right\}$ pass through the origin. Then $y(1)$ is equal to :

(A) $\exp\left(\frac{1-\pi}{4\sqrt{2}}\right)$ (B) $\exp\left(\frac{4-\pi}{4\sqrt{2}}\right)$

(C) $\exp\left(\frac{4+\pi}{4\sqrt{2}}\right)$ (D) $\exp\left(\frac{\pi-4}{4\sqrt{2}}\right)$

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Q68. JEE Main 2023 (Online) 29th January Evening Shift

Let $y = y(x)$ be the solution of the differential equation $x\log_e x\frac{dy}{dx} + y = x^2\log_e x, (x > 1)$. If $y(2) = 2$, then $y(e)$ is equal to

(A) $\frac{1+e^2}{2}$ (B) $\frac{1+e^2}{4}$ (C) $\frac{2+e^2}{2}$ (D) $\frac{4+e^2}{4}$

Open solution

Q69. JEE Main 2023 (Online) 29th January Morning Shift

Let $y = f(x)$ be the solution of the differential equation $y(x+1)dx - x^2dy = 0, y(1) = e$. Then $\lim\limits_{x\to 0^+} f(x)$ is equal to

(A) e^2 (B) 0 (C) $\frac{1}{e^2}$ (D) $\frac{1}{e}$

Open solution

Q70. JEE Main 2023 (Online) 25th January Evening Shift

Let $y = y(t)$ be a solution of the differential equation $\frac{dy}{dt} + \alpha y = \gamma e^{-\beta t}$ where, $\alpha > 0, \beta > 0$ and $\gamma > 0$. Then $\lim\limits_{t\to\infty} y(t)$

(A) is 0 (B) is 1 (C) is -1 (D) does not exist

Open solution

Q71. JEE Main 2023 (Online) 25th January Morning Shift

Let $y = y(x)$ be the solution curve of the differential equation $\frac{dy}{dx} = \frac{y}{x}\left(1 + xy^2(1+\log_e x)\right), x > 0, y(1) = 3$. Then $\frac{y^2(x)}{9}$ is equal to :

(A) $\frac{x^2}{5-2x^3(2+\log_e x^3)}$ (B) $\frac{x^2}{3x^3(1+\log_e x^2)-2}$

(C) $\frac{x^2}{7-3x^3(2+\log_e x^2)}$ (D) $\frac{x^2}{2x^3(2+\log_e x^3)-3}$

Open solution

Q72. JEE Main 2023 (Online) 24th January Evening Shift

Let $y = y(x)$ be the solution of the differential equation $(x^2 - 3y^2)dx + 3xy\,dy = 0, y(1) = 1$. Then $6y^2(e)$ is equal to

(A) $\frac{3}{2}e^2$ (B) $3e^2$ (C) e^2 (D) $2e^2$

Open solution

Q73. JEE Main 2023 (Online) 24th January Morning Shift

Let $y = y(x)$ be the solution of the differential equation $x^3dy + (xy - 1)dx = 0, x > 0, y\left(\frac{1}{2}\right) = 3 - e$. Then y (1) is equal to

(A) $2 - e$ (B) 3 (C) 1 (D) e

Open solution

Q74. JEE Main 2022 (Online) 30th July Evening Shift

If $y = y(x)$ satisfies the differential equation $x\,dy + \left(x\tan\left(\frac{y}{x}\right) - y\right)dx = 0, x > 0$, and $y(1) = \frac{\pi}{2}$, then a value of $y(\sqrt{2})$ is

(A) $\frac{\pi}{3\sqrt{2}}$ (B) $\frac{3}{2\sqrt{2}}$ (C) $\frac{1}{2\sqrt{2}}$ (D) $\frac{3\pi}{2\sqrt{2}}$

Open solution

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Q75. JEE Main 2022 (Online) 30th July Evening Shift

Let the area enclosed by the solution curve $y = y(x)$ of the differential equation $\frac{dy}{dx} + \frac{x-2}{y-b} = 2$, $b > 0$, $y(5) = 0$ is 13π. Let $y = y(x)$ intersect the y-axis at the points P and Q. If the tangents to $y = y(x)$ at P and Q meet at T, then the area of ΔPTQ is

(A) 13.5 (B) 13 (C) 14.5 (D) 15

Open solution

Q76. JEE Main 2022 (Online) 29th July Evening Shift

If the solution curve of the differential equation $\frac{dy}{dx} = \frac{x+y-2}{x-y}$ passes through the points $(2, 1)$ and $(k+1, 2), k > 0$, then

(A) $2\tan^{-1}\left(\frac{1}{k}\right) = \log_e\left(k^2+1\right)$ (B) $\tan^{-1}\left(\frac{1}{k}\right) = \log_e\left(k^2+1\right)$

(C) $2\tan^{-1}\left(\frac{1}{k+1}\right) = \log_e\left(k^2+2k+2\right)$ (D) $2\tan^{-1}\left(\frac{1}{k}\right) = \log_e\left(\frac{k^2+1}{k^2}\right)$

Open solution

Q77. JEE Main 2022 (Online) 29th July Evening Shift

Let $y = y(x)$ be the solution curve of the differential equation $\frac{dy}{dx} + \left(\frac{2x^2+11x+13}{x^3+6x^2+11x+6}\right)y = \frac{(x+3)}{x+1}, x > -1$, which passes through the point $(0, 1)$. Then $y(1)$ is equal to :

(A) $\frac{1}{2}$ (B) $\frac{3}{2}$ (C) $\frac{5}{2}$ (D) $\frac{7}{2}$

Open solution

Q78. JEE Main 2022 (Online) 29th July Morning Shift

Let the solution curve $y = y(x)$ of the differential equation $\left(1+e^{2x}\right)\left(\frac{dy}{dx} + y\right) = 1$ pass through the point $\left(0, \frac{\pi}{2}\right)$. Then, $\lim\limits_{x\to\infty} e^x y(x)$ is equal to :

(A) $\frac{\pi}{4}$ (B) $\frac{3\pi}{4}$ (C) $\frac{\pi}{2}$ (D) $\frac{3\pi}{2}$

Open solution

Q79. JEE Main 2022 (Online) 28th July Evening Shift

Let $y = y(x)$ be the solution curve of the differential equation $\frac{dy}{dx} + \frac{1}{x^2-1}y = \left(\frac{x-1}{x+1}\right)^{1/2}, x > 1$ passing through the point $\left(2, \sqrt{\frac{1}{3}}\right)$. Then $\sqrt{7}\,y(8)$ is equal to :

(A) $11 + 6\log_e 3$ (B) 19 (C) $12 - 2\log_e 3$ (D) $19 - 6\log_e 3$

Open solution

Q80. JEE Main 2022 (Online) 28th July Morning Shift

Let the solution curve of the differential equation $x\,dy = \left(\sqrt{x^2+y^2} + y\right)dx, x > 0$, intersect the line $x = 1$ at $y = 0$ and the line $x = 2$ at $y = \alpha$. Then the value of α is :

(A) $\frac{1}{2}$ (B) $\frac{3}{2}$ (C) $-\frac{3}{2}$ (D) $\frac{5}{2}$

Open solution

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Q81. JEE Main 2022 (Online) 28th July Morning Shift

If $y = y(x), x \in (0, \pi/2)$ be the solution curve of the differential equation

$$(\sin^2 2x)\frac{dy}{dx} + (8\sin^2 2x + 2\sin 4x)y = 2e^{-4x}(2\sin 2x + \cos 2x),$$

with $y(\pi/4) = e^{-\pi}$, then $y(\pi/6)$ is equal to :

(A) $\frac{2}{\sqrt{3}}e^{-2\pi/3}$ (B) $\frac{2}{\sqrt{3}}e^{2\pi/3}$

(C) $\frac{1}{\sqrt{3}}e^{-2\pi/3}$ (D) $\frac{1}{\sqrt{3}}e^{2\pi/3}$

Open solution

Q82. JEE Main 2022 (Online) 27th July Morning Shift

Let $y = y_1(x)$ and $y = y_2(x)$ be two distinct solutions of the differential equation $\frac{dy}{dx} = x + y$, with $y_1(0) = 0$ and $y_2(0) = 1$ respectively. Then, the number of points of intersection of $y = y_1(x)$ and $y = y_2(x)$ is

(A) 0 (B) 1 (C) 2 (D) 3

Open solution

Q83. JEE Main 2022 (Online) 26th July Evening Shift

Let the solution curve $y = f(x)$ of the differential equation $\frac{dy}{dx} + \frac{xy}{x^2-1} = \frac{x^4+2x}{\sqrt{1-x^2}}, x \in (-1, 1)$ pass through the origin. Then $\int_{-\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} f(x)dx$ is equal to

(A) $\frac{\pi}{3} - \frac{1}{4}$ (B) $\frac{\pi}{3} - \frac{\sqrt{3}}{4}$ (C) $\frac{\pi}{6} - \frac{\sqrt{3}}{4}$ (D) $\frac{\pi}{6} - \frac{\sqrt{3}}{2}$

Open solution

Q84. JEE Main 2022 (Online) 26th July Morning Shift

If $\frac{dy}{dx} + 2y\tan x = \sin x$, $0 < x < \frac{\pi}{2}$ and $y\left(\frac{\pi}{3}\right) = 0$, then the maximum value of $y(x)$ is :

(A) $\frac{1}{8}$ (B) $\frac{3}{4}$ (C) $\frac{1}{4}$ (D) $\frac{3}{8}$

Open solution

Q85. JEE Main 2022 (Online) 25th July Morning Shift

The slope of the tangent to a curve $C : y = y(x)$ at any point (x, y) on it is $\frac{2e^{2x}-6e^{-x}+9}{2+9e^{-2x}}$. If C passes through the points $\left(0, \frac{1}{2} + \frac{\pi}{2\sqrt{2}}\right)$ and $\left(\alpha, \frac{1}{2}e^{2\alpha}\right)$, then e^{α} is equal to :

(A) $\frac{3+\sqrt{2}}{3-\sqrt{2}}$ (B) $\frac{3}{\sqrt{2}}\left(\frac{3+\sqrt{2}}{3-\sqrt{2}}\right)$

(C) $\frac{1}{\sqrt{2}}\left(\frac{\sqrt{2}+1}{\sqrt{2}-1}\right)$ (D) $\frac{\sqrt{2}+1}{\sqrt{2}-1}$

Open solution

Q86. JEE Main 2022 (Online) 25th July Morning Shift

The general solution of the differential equation $(x - y^2)dx + y(5x + y^2)dy = 0$ is :

(A) $(y^2 + x)^4 = C\left|(y^2 + 2x)^3\right|$ (B) $(y^2 + 2x)^4 = C\left|(y^2 + x)^3\right|$

(C) $\left|(y^2 + x)^3\right| = C(2y^2 + x)^4$ (D) $\left|(y^2 + 2x)^3\right| = C(2y^2 + x)^4$

Open solution

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Q87. JEE Main 2022 (Online) 29th June Evening Shift

If y = y(x) is the solution of the differential equation $\left(1+e^{2x}\right)\frac{dy}{dx}+2\left(1+y^2\right)e^x=0$ and y (0) = 0, then $6\left(y'(0)+\left(y\left(\log_e\sqrt{3}\right)\right)^2\right)$ is equal to

(A) 2 (B) -2 (C) -4 (D) -1

Open solution

Q88. JEE Main 2022 (Online) 29th June Morning Shift

Let the solution curve of the differential equation

$x\frac{dy}{dx}-y=\sqrt{y^2+16x^2}$, $y(1)=3$ be $y=y(x)$. Then y(2) is equal to:

(A) 15 (B) 11 (C) 13 (D) 17

Open solution

Q89. JEE Main 2022 (Online) 28th June Evening Shift

Let x = x(y) be the solution of the differential equation

$2y\,e^{x/y^2}dx+\left(y^2-4xe^{x/y^2}\right)dy=0$ such that x(1) = 0. Then, x(e) is equal to :

(A) $e\log_e(2)$ (B) $-e\log_e(2)$ (C) $e^2\log_e(2)$ (D) $-e^2\log_e(2)$

Open solution

Q90. JEE Main 2022 (Online) 28th June Evening Shift

Let the slope of the tangent to a curve y = f(x) at (x, y) be given by $2\tan x(\cos x-y)$. If the curve passes through the point $\left(\frac{\pi}{4},0\right)$, then the value of $\int_0^{\pi/2} y\,dx$ is equal to :

(A) $(2-\sqrt{2})+\frac{\pi}{\sqrt{2}}$ (B) $2-\frac{\pi}{\sqrt{2}}$

(C) $(2+\sqrt{2})+\frac{\pi}{\sqrt{2}}$ (D) $2+\frac{\pi}{\sqrt{2}}$

Open solution

Q91. JEE Main 2022 (Online) 28th June Morning Shift

Let the solution curve $y=y(x)$ of the differential equation

$\left[\frac{x}{\sqrt{x^2-y^2}}+e^{\frac{y}{x}}\right]x\frac{dy}{dx}=x+\left[\frac{x}{\sqrt{x^2-y^2}}+e^{\frac{y}{x}}\right]y$

pass through the points (1, 0) and $(2\alpha, \alpha)$, $\alpha > 0$. Then α is equal to

(A) $\frac{1}{2}\exp\left(\frac{\pi}{6}+\sqrt{e}-1\right)$ (B) $\frac{1}{2}\exp\left(\frac{\pi}{6}+e-1\right)$

(C) $\exp\left(\frac{\pi}{6}+\sqrt{e}+1\right)$ (D) $2\exp\left(\frac{\pi}{3}+\sqrt{e}-1\right)$

Open solution

Q92. JEE Main 2022 (Online) 28th June Morning Shift

Let y = y(x) be the solution of the differential equation $x(1-x^2)\frac{dy}{dx}+(3x^2y-y-4x^3)=0$, $x>1$, with $y(2)=-2$. Then y(3) is equal to :

(A) -18 (B) -12 (C) -6 (D) -3

Open solution

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Q93. JEE Main 2022 (Online) 27th June Evening Shift

If the solution curve of the differential equation

$((\tan^{-1}y) - x)dy = (1 + y^2)dx$ passes through the point (1, 0), then the abscissa of the point on the curve whose ordinate is tan(1), is

Open solution

(A) 2e (B) $\frac{2}{e}$ (C) 2 (D) $\frac{1}{e}$

Q94. JEE Main 2022 (Online) 27th June Morning Shift

If $\frac{dy}{dx} + \frac{2^{x-y}(2^y-1)}{2^x-1} = 0$, x, y > 0, y(1) = 1, then y(2) is equal to :

Open solution

(A) $2 + \log_2 3$ (B) $2 + \log_3 2$ (C) $2 - \log_3 2$ (D) $2 - \log_2 3$

Q95. JEE Main 2022 (Online) 26th June Evening Shift

If $y = y(x)$ is the solution of the differential equation

$x\frac{dy}{dx} + 2y = x\,e^x, y(1) = 0$ then the local maximum value

Open solution

of the function $z(x) = x^2y(x) - e^x$, $x \in R$ is :

(A) 1 − e (B) 0 (C) $\frac{1}{2}$ (D) $\frac{4}{e} - e$

Q96. JEE Main 2022 (Online) 26th June Evening Shift

If the solution of the differential equation

$\frac{dy}{dx} + e^x\left(x^2 - 2\right)y = \left(x^2 - 2x\right)\left(x^2 - 2\right)e^{2x}$ satisfies $y(0) = 0$, then the value of y(2) is _____.

Open solution

(A) −1 (B) 1 (C) 0 (D) e

Q97. JEE Main 2022 (Online) 25th June Evening Shift

If $y = y(x)$ is the solution of the differential equation

$2x^2\frac{dy}{dx} - 2xy + 3y^2 = 0$ such that $y(e) = \frac{e}{3}$, then y(1) is equal to :

Open solution

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $\frac{3}{2}$ (D) 3

Q98. JEE Main 2022 (Online) 25th June Morning Shift

Let $y = y(x)$ be the solution of the differential equation $(x + 1)y' - y = e^{3x}(x + 1)^2$, with $y(0) = \frac{1}{3}$. Then, the point $x = -\frac{4}{3}$ for the curve $y = y(x)$ is :

Open solution

(A) not a critical point (B) a point of local minima

(C) a point of local maxima (D) a point of inflection

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Q99. JEE Main 2022 (Online) 25th June Morning Shift

If the solution curve $y = y(x)$ of the differential equation $y^2dx + (x^2 - xy + y^2)dy = 0$, which passes through the point (1, 1) and intersects the line $y = \sqrt{3}x$ at the point $(\alpha, \sqrt{3}\alpha)$, then value of $\log_e(\sqrt{3}\alpha)$ is equal to :

(A) $\frac{\pi}{3}$ (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{12}$ (D) $\frac{\pi}{6}$

Open solution

Q100. JEE Main 2022 (Online) 24th June Morning Shift

If x = x(y) is the solution of the differential equation

$y\frac{dx}{dy} = 2x + y^3(y+1)e^y,\ x(1) = 0$; then x(e) is equal to :

(A) $e^3(e^e - 1)$ (B) $e^e(e^3 - 1)$ (C) $e^2(e^e + 1)$ (D) $e^e(e^2 - 1)$

Open solution

Q101. JEE Main 2021 (Online) 1st September Evening Shift

If y = y(x) is the solution curve of the differential equation $x^2dy + \left(y - \frac{1}{x}\right)dx = 0$; x > 0 and y(1) = 1, then $y\left(\frac{1}{2}\right)$ is equal to :

(A) $\frac{3}{2} - \frac{1}{\sqrt{e}}$ (B) $3 + \frac{1}{\sqrt{e}}$

(C) 3 + e (D) 3 − e

Open solution

Q102. JEE Main 2021 (Online) 31st August Evening Shift

If $\frac{dy}{dx} = \frac{2^xy + 2^y.2^x}{2^x + 2^{x+y}\log_e 2}$, y(0) = 0, then for y = 1, the value of x lies in the interval :

(A) (1, 2) (B) $\left(\frac{1}{2}, 1\right]$ (C) (2, 3) (D) $\left(0, \frac{1}{2}\right]$

Open solution

Q103. JEE Main 2021 (Online) 31st August Evening Shift

If $y\frac{dy}{dx} = x\left[\frac{y^2}{x^2} + \frac{\phi\left(\frac{y^2}{x^2}\right)}{\phi'\left(\frac{y^2}{x^2}\right)}\right]$, x > 0, ϕ > 0, and y(1) = −1, then $\phi\left(\frac{y^2}{4}\right)$ is equal to :

(A) 4 ϕ (2) (B) 4ϕ (1) (C) 2 ϕ (1) (D) ϕ (1)

Open solution

Q104. JEE Main 2021 (Online) 31st August Morning Shift

If $\frac{dy}{dx} = \frac{2^{x+y} - 2^x}{2^y}$, y(0) = 1, then y(1) is equal to :

(A) $\log_2(2 + e)$ (B) $\log_2(1 + e)$ (C) $\log_2(2e)$ (D) $\log_2(1 + e^2)$

Open solution

Q105. JEE Main 2021 (Online) 27th August Evening Shift

If the solution curve of the differential equation $(2x - 10y^3)dy + ydx = 0$, passes through the points (0, 1) and (2, β), then β is a root of the equation :

(A) $y^5 - 2y - 2 = 0$ (B) $2y^5 - 2y - 1 = 0$ (C) $2y^5 - y^2 - 2 = 0$ (D) $y^5 - y^2 - 1 = 0$

Open solution

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Q106. JEE Main 2021 (Online) 27th August Morning Shift

Let y = y(x) be the solution of the differential equation

$\frac{dy}{dx} = 2(y + 2\sin x - 5)x - 2\cos x$ such that y(0) = 7. Then y(π) is equal to :

Open solution

(A) $2e^{\pi^2} + 5$ (B) $e^{\pi^2} + 5$ (C) $3e^{\pi^2} + 5$ (D) $7e^{\pi^2} + 5$

Q107. JEE Main 2021 (Online) 27th August Morning Shift

Let us consider a curve, y = f(x) passing through the point $(-2, 2)$ and the slope of the tangent to the curve at any point (x, f(x)) is given by f(x) + xf'(x) = x^2. Then :

Open solution

(A) $x^2 + 2xf(x) - 12 = 0$ (B) $x^3 + xf(x) + 12 = 0$

(C) $x^3 - 3xf(x) - 4 = 0$ (D) $x^2 + 2xf(x) + 4 = 0$

Q108. JEE Main 2021 (Online) 26th August Evening Shift

Let y(x) be the solution of the differential equation

Open solution

$2x^2\,dy + (e^y - 2x)dx = 0$, x > 0. If y(e) = 1, then y(1) is equal to :

(A) 0 (B) 2 (C) $\log_e 2$ (D) $\log_e (2e)$

Q109. JEE Main 2021 (Online) 26th August Morning Shift

Let y = y(x) be a solution curve of the differential equation $(y+1)\tan^2 x\,dx + \tan x\,dy + y\,dx = 0$, $x \in \left(0, \frac{\pi}{2}\right)$. If $\lim\limits_{x\to 0+} xy(x) = 1$, then the value of $y\left(\frac{\pi}{4}\right)$ is :

Open solution

(A) $-\frac{\pi}{4}$ (B) $\frac{\pi}{4} - 1$ (C) $\frac{\pi}{4} + 1$ (D) $\frac{\pi}{4}$

Q110. JEE Main 2021 (Online) 27th July Evening Shift

Let y = y(x) be the solution of the differential

Open solution

equation $(x - x^3)dy = (y + yx^2 - 3x^4)dx$, x > 2. If y(3) = 3, then y(4) is equal to :

(A) 4 (B) 12 (C) 8 (D) 16

Q111. JEE Main 2021 (Online) 27th July Morning Shift

Let y = y(x) be solution of the differential equation

Open solution

$\log_e\left(\frac{dy}{dx}\right) = 3x + 4y$, with y(0) = 0.

If $y\left(-\frac{2}{3}\log_e 2\right) = \alpha\log_e 2$, then the value of α is equal to :

(A) $-\frac{1}{4}$ (B) $\frac{1}{4}$ (C) 2 (D) $-\frac{1}{2}$

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Q112. JEE Main 2021 (Online) 25th July Evening Shift

Let y = y(x) be the solution of the differential

equation xdy = (y + x^3 cosx)dx with y(π) = 0, then $y\left(\frac{\pi}{2}\right)$ is equal to :

Open solution

(A) $\frac{\pi^2}{4}+\frac{\pi}{2}$ (B) $\frac{\pi^2}{2}+\frac{\pi}{4}$

(C) $\frac{\pi^2}{2}-\frac{\pi}{4}$ (D) $\frac{\pi^4}{4}-\frac{\pi}{2}$

Q113. JEE Main 2021 (Online) 25th July Morning Shift

Let y = y(x) be the solution of the differential equation $\frac{dy}{dx}=1+xe^{y-x}, -\sqrt{2}<x<\sqrt{2}, y(0)=0$

then, the minimum value of $y(x), x\in\left(-\sqrt{2},\sqrt{2}\right)$ is equal to :

Open solution

(A) $\left(2-\sqrt{3}\right)-\log_e 2$ (B) $\left(2+\sqrt{3}\right)+\log_e 2$

(C) $\left(1+\sqrt{3}\right)-\log_e\left(\sqrt{3}-1\right)$ (D) $\left(1-\sqrt{3}\right)-\log_e\left(\sqrt{3}-1\right)$

Q114. JEE Main 2021 (Online) 22th July Evening Shift

Let y = y(x) be the solution of the differential equation $\text{cosec}^2 x\,dy+2dx=(1+y\cos 2x)\text{cosec}^2 x\,dx$, with $y\left(\frac{\pi}{4}\right)=0$. Then, the value of $(y(0)+1)^2$ is equal to :

Open solution

(A) $e^{1/2}$ (B) $e^{-1/2}$ (C) e^{-1} (D) e

Q115. JEE Main 2021 (Online) 20th July Evening Shift

Let y = y(x) satisfies the equation $\frac{dy}{dx}-|A|=0$, for all x > 0, where $A=\begin{bmatrix} y & \sin x & 1 \\ 0 & -1 & 1 \\ 2 & 0 & \frac{1}{x}\end{bmatrix}$. If $y(\pi)=\pi+2$, then the value of $y\left(\frac{\pi}{2}\right)$ is :

Open solution

(A) $\frac{\pi}{2}+\frac{4}{\pi}$ (B) $\frac{\pi}{2}-\frac{1}{\pi}$

(C) $\frac{3\pi}{2}-\frac{1}{\pi}$ (D) $\frac{\pi}{2}-\frac{4}{\pi}$

Q116. JEE Main 2021 (Online) 20th July Morning Shift

Let y = y(x) be the solution of the differential equation $x\tan\left(\frac{y}{x}\right)dy=\left(y\tan\left(\frac{y}{x}\right)-x\right)dx, -1\le x\le 1$, $y\left(\frac{1}{2}\right)=\frac{\pi}{6}$. Then the area of the region bounded by the curves x = 0, $x=\frac{1}{\sqrt{2}}$ and y = y(x) in the upper half plane is :

Open solution

(A) $\frac{1}{8}(\pi-1)$ (B) $\frac{1}{12}(\pi-3)$ (C) $\frac{1}{4}(\pi-2)$ (D) $\frac{1}{6}(\pi-1)$

Q117. JEE Main 2021 (Online) 20th July Morning Shift

Let y = y(x) be the solution of the differential equation $e^x\sqrt{1-y^2}dx+\left(\frac{y}{x}\right)dy=0$, y(1) = -1. Then the value of $(y(3))^2$ is equal to :

Open solution

(A) $1-4e^3$ (B) $1-4e^6$ (C) $1+4e^3$ (D) $1+4e^6$

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Q118. JEE Main 2021 (Online) 18th March Evening Shift

Open solution

Let y = y(x) be the solution of the differential equation

$\frac{dy}{dx} = (y+1)\left((y+1)e^{x^2/2} - x\right)$, 0 < x < 2.1, with y(2) = 0. Then the value of $\frac{dy}{dx}$ at x = 1 is equal to :

(A) $\frac{e^{5/2}}{(1+e^2)^2}$

(B) $\frac{5e^{1/2}}{(e^2+1)^2}$

(C) $-\frac{2e^2}{(1+e^2)^2}$

(D) $\frac{-e^{3/2}}{(e^2+1)^2}$

Q119. JEE Main 2021 (Online) 17th March Evening Shift

Open solution

If the curve y = y(x) is the solution of the differential equation

$2(x^2 + x^{5/4})dy - y(x + x^{1/4})dx = 2x^{9/4}dx$, x > 0 which

passes through the point $\left(1, 1 - \frac{4}{3}\log_e 2\right)$, then the value of y(16) is equal to :

(A) $4\left(\frac{31}{3} - \frac{8}{3}\log_e 3\right)$

(B) $\left(\frac{31}{3} - \frac{8}{3}\log_e 3\right)$

(C) $\left(\frac{31}{3} + \frac{8}{3}\log_e 3\right)$

(D) $4\left(\frac{31}{3} + \frac{8}{3}\log_e 3\right)$

Q120. JEE Main 2021 (Online) 17th March Evening Shift

Open solution

Let y = y(x) be the solution of the differential equation

$\cos x(3\sin x + \cos x + 3)dy = (1 + y\sin x(3\sin x + \cos x + 3))dx, 0 \le x \le \frac{\pi}{2}, y(0) = 0$. Then, $y\left(\frac{\pi}{3}\right)$ is equal to :

(A) $2\log_e\left(\frac{\sqrt{3}+7}{2}\right)$

(B) $2\log_e\left(\frac{3\sqrt{3}-8}{4}\right)$

(C) $2\log_e\left(\frac{2\sqrt{3}+10}{11}\right)$

(D) $2\log_e\left(\frac{2\sqrt{3}+9}{6}\right)$

Q121. JEE Main 2021 (Online) 17th March Morning Shift

Open solution

Which of the following is true for y(x) that satisfies the differential equation

$\frac{dy}{dx}$ = xy − 1 + x − y; y(0) = 0 :

(A) y(1) = 1

(B) $y(1) = e^{-\frac{1}{2}} - 1$

(C) $y(1) = e^{\frac{1}{2}} - e^{-\frac{1}{2}}$

(D) $y(1) = e^{\frac{1}{2}} - 1$

Q122. JEE Main 2021 (Online) 16th March Evening Shift

Open solution

If y = y(x) is the solution of the differential equation

$\frac{dy}{dx}$ + (tan x) y = sin x, $0 \le x \le \frac{\pi}{3}$, with y(0) = 0, then $y\left(\frac{\pi}{4}\right)$ equal to :

(A) $\frac{1}{2}\log_e 2$

(B) $\left(\frac{1}{2\sqrt{2}}\right)\log_e 2$

(C) $\log_e 2$

(D) $\frac{1}{4}\log_e 2$

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Q123. JEE Main 2021 (Online) 16th March Evening Shift

Let C_1 be the curve obtained by the solution of differential equation

$2xy\frac{dy}{dx} = y^2 - x^2, x > 0$. Let the curve C_2 be the

solution of $\frac{2xy}{x^2-y^2} = \frac{dy}{dx}$. If both the curves pass through (1, 1), then the area enclosed by the curves C_1 and C_2 is equal to :

Open solution

(A) $\frac{\pi}{4}$ + 1 (B) π + 1 (C) $\pi - 1$ (D) $\frac{\pi}{2} - 1$

Q124. JEE Main 2021 (Online) 16th March Morning Shift

If y = y(x) is the solution of the differential equation,

$\frac{dy}{dx} + 2y\tan x = \sin x, y\left(\frac{\pi}{3}\right) = 0$, then the maximum value of the function y(x) over R is equal to:

Open solution

(A) $\frac{1}{8}$ (B) 8 (C) $-\frac{15}{4}$ (D) $\frac{1}{2}$

Q125. JEE Main 2021 (Online) 26th February Morning Shift

The rate of growth of bacteria in a culture is proportional to the number of bacteria present and the bacteria count is 1000 at initial time t = 0. The number of bacteria is increased by 20% in 2 hours. If the population of bacteria is 2000 after $\frac{k}{\log_e\left(\frac{6}{5}\right)}$ hours, then $\left(\frac{k}{\log_e 2}\right)^2$ is equal to :

Open solution

(A) 16 (B) 8 (C) 2 (D) 4

Q126. JEE Main 2021 (Online) 25th February Morning Shift

If a curve passes through the origin and the slope of the tangent to it at any point (x, y) is $\frac{x^2-4x+y+8}{x-2}$, then this curve also passes through the point :

Open solution

(A) (4, 4) (B) (5, 5) (C) (5, 4) (D) (4, 5)

Q127. JEE Main 2021 (Online) 24th February Morning Shift

The population P = P(t) at time 't' of a certain species follows the differential equation

$\frac{dP}{dt}$ = 0.5P – 450. If P(0) = 850, then the time at which population becomes zero is :

Open solution

(A) $\log_e 18$ (B) $\frac{1}{2}\log_e 18$ (C) $2\log_e 18$ (D) $\log_e 9$

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Q128. JEE Main 2020 (Online) 6th September Morning Slot

The general solution of the differential equation

$\sqrt{1+x^2+y^2+x^2y^2}$ + xy$\frac{dy}{dx}$ = 0 is :

(where C is a constant of integration)

(A) $\sqrt{1+y^2}+\sqrt{1+x^2}=\frac{1}{2}\log_e\left(\frac{\sqrt{1+x^2}-1}{\sqrt{1+x^2}+1}\right)+C$

(B) $\sqrt{1+y^2}-\sqrt{1+x^2}=\frac{1}{2}\log_e\left(\frac{\sqrt{1+x^2}-1}{\sqrt{1+x^2}+1}\right)+C$

(C) $\sqrt{1+y^2}+\sqrt{1+x^2}=\frac{1}{2}\log_e\left(\frac{\sqrt{1+x^2}+1}{\sqrt{1+x^2}-1}\right)+C$

(D) $\sqrt{1+y^2}-\sqrt{1+x^2}=\frac{1}{2}\log_e\left(\frac{\sqrt{1+x^2}+1}{\sqrt{1+x^2}-1}\right)+C$

Open solution

Q129. JEE Main 2020 (Online) 5th September Evening Slot

Let y = y(x) be the solution of the differential equation

cosx$\frac{dy}{dx}$ + 2ysinx = sin2x, x $\in \left(0, \frac{\pi}{2}\right)$.

If y$\left(\frac{\pi}{3}\right)$ = 0, then y$\left(\frac{\pi}{4}\right)$ is equal to :

(A) $\frac{1}{\sqrt{2}}-1$ (B) $\sqrt{2}-2$ (C) $2-\sqrt{2}$ (D) $2+\sqrt{2}$

Open solution

Q130. JEE Main 2020 (Online) 5th September Morning Slot

If y = y(x) is the solution of the differential

equation $\frac{5+e^x}{2+y}.\frac{dy}{dx}+e^x=0$ satisfying y(0) = 1, then a value of y($\log_e$13) is :

(A) -1 (B) 1 (C) 0 (D) 2

Open solution

Q131. JEE Main 2020 (Online) 4th September Evening Slot

The solution of the differential equation

$\frac{dy}{dx}-\frac{y+3x}{\log_e(y+3x)}+3=0$ is:

(where c is a constant of integration)

(A) $x-\frac{1}{2}(\log_e(y+3x))^2=C$ (B) $y+3x-\frac{1}{2}(\log_e x)^2=C$

(C) x − $\log_e$(y+3x) = C (D) x − 2$\log_e$(y+3x) = C

Open solution

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Q132. JEE Main 2020 (Online) 4th September Morning Slot

Open solution

Let y = y(x) be the solution of the differential equation,
xy'- y = x^2(xcosx + sinx), x > 0. if y (π) = π then
$y''\left(\frac{\pi}{2}\right) + y\left(\frac{\pi}{2}\right)$ is equal to :

(A) $1 + \frac{\pi}{2}$

(B) $2 + \frac{\pi}{2} + \frac{\pi^2}{4}$

(C) $2 + \frac{\pi}{2}$

(D) $1 + \frac{\pi}{2} + \frac{\pi^2}{4}$

Q133. JEE Main 2020 (Online) 3rd September Evening Slot

Open solution

If x^3dy + xy dx = x^2dy + 2y dx; y(2) = e and
x > 1, then y(4) is equal to :

(A) $\frac{\sqrt{e}}{2}$

(B) $\frac{1}{2} + \sqrt{e}$

(C) $\frac{3}{2} + \sqrt{e}$

(D) $\frac{3}{2}\sqrt{e}$

Q134. JEE Main 2020 (Online) 3rd September Morning Slot

Open solution

The solution curve of the differential equation,

$(1 + e^{-x})(1 + y^2)\frac{dy}{dx} = y^2$,

which passes through the point (0, 1), is :

(A) $y^2 + 1 = y\left(\log_e\left(\frac{1+e^{-x}}{2}\right) + 2\right)$

(B) $y^2 + 1 = y\left(\log_e\left(\frac{1+e^{x}}{2}\right) + 2\right)$

(C) $y^2 = 1 + y\log_e\left(\frac{1+e^{-x}}{2}\right)$

(D) $y^2 = 1 + y\log_e\left(\frac{1+e^{x}}{2}\right)$

Q135. JEE Main 2020 (Online) 2nd September Evening Slot

Open solution

If a curve y = f(x), passing through the point (1, 2), is the solution of the differential equation,
$2x^2dy = (2xy + y^2)dx$, then $f\left(\frac{1}{2}\right)$ is equal to :

(A) $\frac{1}{1-\log_e 2}$

(B) $\frac{1}{1+\log_e 2}$

(C) $\frac{-1}{1+\log_e 2}$

(D) $1 + \log_e 2$

Q136. JEE Main 2020 (Online) 2nd September Morning Slot

Open solution

Let y = y(x) be the solution of the differential equation,
$\frac{2+\sin x}{y+1}\cdot\frac{dy}{dx} = -\cos x$, y > 0,y(0) = 1.
If y(π) = a and $\frac{dy}{dx}$ at x = π is b, then the ordered pair (a, b) is equal to :

(A) (2, 1)

(B) $\left(2, \frac{3}{2}\right)$

(C) (1, -1)

(D) (1, 1)

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Q137. JEE Main 2020 (Online) 9th January Evening Slot

If $\frac{dy}{dx} = \frac{xy}{x^2+y^2}$; y(1) = 1; then a value of x satisfying y(x) = e is :

(A) $\sqrt{2}e$ (B) $\frac{1}{2}\sqrt{3}e$ (C) $\frac{e}{\sqrt{2}}$ (D) $\sqrt{3}e$

Open solution

Q138. JEE Main 2020 (Online) 8th January Morning Slot

Let y = y(x) be a solution of the differential equation,

$\sqrt{1-x^2}\frac{dy}{dx} + \sqrt{1-y^2} = 0$, |x| < 1.

If $y\left(\frac{1}{2}\right) = \frac{\sqrt{3}}{2}$, then $y\left(-\frac{1}{\sqrt{2}}\right)$ is equal to :

(A) $-\frac{\sqrt{3}}{2}$ (B) None of those (C) $\frac{1}{\sqrt{2}}$ (D) $-\frac{1}{\sqrt{2}}$

Open solution

Q139. JEE Main 2020 (Online) 7th January Evening Slot

Let y = y(x) be the solution curve of the differential equation,

$(y^2 - x)\frac{dy}{dx} = 1$, satisfying y(0) = 1. This curve intersects the x-axis at a point whose abscissa is :

(A) 2 + e (B) -e (C) 2 (D) 2 - e

Open solution

Q140. JEE Main 2020 (Online) 7th January Morning Slot

If y = y(x) is the solution of the differential equation, $e^y\left(\frac{dy}{dx} - 1\right) = e^x$ such that y(0) = 0, then y(1) is equal to:

(A) $2 + \log_e 2$ (B) $\log_e 2$ (C) $1 + \log_e 2$ (D) 2e

Open solution

Q141. JEE Main 2019 (Online) 12th April Evening Slot

The general solution of the differential equation $(y^2 - x^3)dx - xydy = 0$ $(x \neq 0)$ is : (where c is a constant of integration)

(A) $y^2 + 2x^3 + cx^2 = 0$ (B) $y^2 + 2x^2 + cx^3 = 0$

(C) $y^2 - 2x + cx^3 = 0$ (D) $y^2 - 2x^3 + cx^2 = 0$

Open solution

Q142. JEE Main 2019 (Online) 12th April Morning Slot

Consider the differential equation, $y^2dx + \left(x - \frac{1}{y}\right)dy = 0$, If value of y is 1 when x = 1, then the value of x for which y = 2, is :

(A) $\frac{3}{2} - \frac{1}{\sqrt{e}}$ (B) $\frac{1}{2} + \frac{1}{\sqrt{e}}$

(C) $\frac{5}{2} + \frac{1}{\sqrt{e}}$ (D) $\frac{3}{2} - \sqrt{e}$

Open solution

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Q143. JEE Main 2019 (Online) 10th April Evening Slot

Let y = y(x) be the solution of the differential equation,
$\frac{dy}{dx} + y\tan x = 2x + x^2\tan x, x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, such that y(0) = 1. Then :

(A) $y\left(\frac{\pi}{4}\right) - y\left(-\frac{\pi}{4}\right) = \sqrt{2}$

(B) $y'\left(\frac{\pi}{4}\right) - y'\left(-\frac{\pi}{4}\right) = \pi - \sqrt{2}$

(C) $y\left(\frac{\pi}{4}\right) + y\left(-\frac{\pi}{4}\right) = \frac{\pi^2}{2} + 2$

(D) $y'\left(\frac{\pi}{4}\right) + y'\left(-\frac{\pi}{4}\right) = -\sqrt{2}$

Open solution

Q144. JEE Main 2019 (Online) 10th April Morning Slot

If y = y(x) is the solution of the differential equation
$\frac{dy}{dx} = (\tan x - y)\sec^2 x, x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$,
such that y (0) = 0, then $y\left(-\frac{\pi}{4}\right)$ is equal to :

(A) $\frac{1}{2} - e$ (B) $e - 2$ (C) $2 + \frac{1}{e}$ (D) $\frac{1}{e} - 2$

Open solution

Q145. JEE Main 2019 (Online) 9th April Evening Slot

If $\cos x\frac{dy}{dx} - y\sin x = 6x$, (0 < x < $\frac{\pi}{2}$)
and $y\left(\frac{\pi}{3}\right)$ = 0 then $y\left(\frac{\pi}{6}\right)$ is equal to :-

(A) $-\frac{\pi^2}{2}$ (B) $-\frac{\pi^2}{4\sqrt{3}}$

(C) $\frac{\pi^2}{2\sqrt{3}}$ (D) $-\frac{\pi^2}{2\sqrt{3}}$

Open solution

Q146. JEE Main 2019 (Online) 9th April Morning Slot

The solution of the differential equation

$x\frac{dy}{dx} + 2y = x^2$ ($x \neq 0$) with y(1) = 1, is :

(A) $y = \frac{4}{5}x^3 + \frac{1}{5x^2}$ (B) $y = \frac{3}{4}x^2 + \frac{1}{4x^2}$

(C) $y = \frac{x^2}{4} + \frac{3}{4x^2}$ (D) $y = \frac{x^3}{5} + \frac{1}{5x^2}$

Open solution

Q147. JEE Main 2019 (Online) 8th April Morning Slot

Let y = y(x) be the solution of the differential equation,

$(x^2+1)^2\frac{dy}{dx} + 2x(x^2+1)y = 1$

such that y(0) = 0. If $\sqrt{a}y(1) = \frac{\pi}{32}$, then the value of 'a' is :

(A) $\frac{1}{2}$ (B) $\frac{1}{16}$ (C) 1 (D) $\frac{1}{4}$

Open solution

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Q148. JEE Main 2019 (Online) 12th January Evening Slot

If a curve passes through the point (1, −2) and has slope of the tangent at any point (x, y) on it as $\frac{x^2-2y}{x}$, then the curve also passes through the point :

(A) (−1, 2) (B) $\left(-\sqrt{2}, 1\right)$ (C) $\left(\sqrt{3}, 0\right)$ (D) (3, 0)

Open solution

Q149. JEE Main 2019 (Online) 12th January Morning Slot

Let y = y(x) be the solution of the differential equation, $x\frac{dy}{dx}$ + y = x $\log_e$ x, (x > 1). If 2y(2) = $\log_e$ 4 − 1, then y(e) is equal to :

(A) $-\frac{e}{2}$ (B) $-\frac{e^2}{2}$ (C) $\frac{e^2}{4}$ (D) $\frac{e}{4}$

Open solution

Q150. JEE Main 2019 (Online) 11th January Evening Slot

The solution of the differential equation,

$\frac{dy}{dx}$ = $(x - y)^2$, when y(1) = 1, is :

(A) $-\log_e\left|\frac{1+x-y}{1-x+y}\right|$ = x + y − 2

(B) $\log_e\left|\frac{2-x}{2-y}\right|$ = x − y

(C) $\log_e\left|\frac{2-y}{2-x}\right|$ = 2(y − 1)

(D) $-\log_e\left|\frac{1-x+y}{1+x-y}\right|$ = 2(x − 1)

Open solution

Q151. JEE Main 2019 (Online) 11th January Morning Slot

If y(x) is the solution of the differential equation $\frac{dy}{dx} + \left(\frac{2x+1}{x}\right)y = e^{-2x},\ x > 0,$ where $y(1) = \frac{1}{2}e^{-2}$, then

(A) y($\log_e$2) = $\log_e$4

(B) y(x) is decreasing in (0, 1)

(C) y($\log_e$2) = $\frac{\log_e 2}{4}$

(D) y(x) is decreasing in $\left(\frac{1}{2}, 1\right)$

Open solution

Q152. JEE Main 2019 (Online) 10th January Evening Slot

The curve amongst the family of curves represented by the differential equation, $(x^2 - y^2)dx + 2xy\,dy = 0$ which passes through (1, 1) is :

(A) a circle with centre on the y-axis

(B) an ellipse with major axis along the y-axis

(C) a circle with centre on the x-axis

(D) a hyperbola with transverse axis along the x-axis

Open solution

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Q153. JEE Main 2019 (Online) 10th January Evening Slot

Let f be a differentiable function such that f '(x) = 7 - $\frac{3}{4}\frac{f(x)}{x}$, (x > 0) and f(1) $\neq$ 4. Then $\lim_{x\to 0^+} xf\left(\frac{1}{x}\right)$:

(A) does not exist
(B) exists and equals $\frac{4}{7}$
(C) exists and equals 4
(D) exists and equals 0

Open solution

Q154. JEE Main 2019 (Online) 10th January Morning Slot

If $\frac{dy}{dx} + \frac{3}{\cos^2 x}y = \frac{1}{\cos^2 x}$, $x \in \left(\frac{-\pi}{3}, \frac{\pi}{3}\right)$ and $y\left(\frac{\pi}{4}\right) = \frac{4}{3}$, then $y\left(-\frac{\pi}{4}\right)$ equals -

(A) $\frac{1}{3} + e^6$
(B) $\frac{1}{3}$
(C) $\frac{1}{3}$ + e^3
(D) $-\frac{4}{3}$

Open solution

Q155. JEE Main 2019 (Online) 9th January Evening Slot

Let f : [0,1] $\rightarrow$ **R** be such that f(xy) = f(x).f(y), for all x, y $\in$ [0, 1], and f(0) $\neq$ 0. If y = y(x) satiesfies the differential equation, $\frac{dy}{dx}$ = f(x) with y(0) = 1, then y$\left(\frac{1}{4}\right)$ + y$\left(\frac{3}{4}\right)$ is equal to :

(A) 3
(B) 4
(C) 2
(D) 5

Open solution

Q156. JEE Main 2019 (Online) 9th January Morning Slot

If y = y(x) is the solution of the differential equation,

x$\frac{dy}{dx}$ + 2y = x^2, satisfying y(1) = 1, then y($\frac{1}{2}$) is equal to :

(A) $\frac{7}{64}$
(B) $\frac{49}{16}$
(C) $\frac{1}{4}$
(D) $\frac{13}{16}$

Open solution

Q157. JEE Main 2018 (Offline)

Let y = y(x) be the solution of the differential equation

$\sin x\frac{dy}{dx} + y\cos x = 4x, x \in (0, \pi)$.

If $y\left(\frac{\pi}{2}\right) = 0$, then $y\left(\frac{\pi}{6}\right)$ is equal to :

(A) $-\frac{4}{9}\pi^2$
(B) $\frac{4}{9\sqrt{3}}\pi^2$
(C) $-\frac{8}{9\sqrt{3}}\pi^2$
(D) $-\frac{8}{9}\pi^2$

Open solution

Q158. JEE Main 2018 (Online) 15th April Evening Slot

The curve satisfying the differential equation, ($x^2 - y^2$) dx + 2xydy = 0 and passing through the point (1, 1) is :

(A) a circle of radius one.
(B) a hyperbola.
(C) an ellipse.
(D) a circle of radius two.

Open solution

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Q159. JEE Main 2018 (Online) 15th April Morning Slot

Let y = y(x) be the solution of the differential equation $\frac{dy}{dx} + 2y = f(x)$,

where $f(x) = \begin{cases} 1, & x \in [0,1] \\ 0, & \text{otherwise} \end{cases}$

If y(0) = 0, then $y\left(\frac{3}{2}\right)$ is :

(A) $\frac{e^2+1}{2e^4}$ (B) $\frac{1}{2e}$ (C) $\frac{e^2-1}{e^3}$ (D) $\frac{e^2-1}{2e^3}$

Open solution

Q160. JEE Main 2017 (Online) 9th April Morning Slot

If $2x = y^{\frac{1}{5}} + y^{-\frac{1}{5}}$ and

$(x^2 - 1)\frac{d^2y}{dx^2} + \lambda x \frac{dy}{dx} + ky = 0$,

then $\lambda + k$ is equal to :

(A) -23 (B) -24 (C) 26 (D) -26

Open solution

Q161. JEE Main 2017 (Online) 8th April Morning Slot

The curve satisfying the differential equation, $ydx - (x + 3y^2)dy = 0$ and passing through the point (1, 1), also passes through the point :

(A) $\left(\frac{1}{4}, -\frac{1}{2}\right)$ (B) $\left(-\frac{1}{3}, \frac{1}{3}\right)$

(C) $\left(\frac{1}{3}, -\frac{1}{3}\right)$ (D) $\left(\frac{1}{4}, \frac{1}{2}\right)$

Open solution

Q162. JEE Main 2017 (Offline)

If $(2 + \sin x)\frac{dy}{dx} + (y + 1)\cos x = 0$ and y(0) = 1,

then $y\left(\frac{\pi}{2}\right)$ is equal to :

(A) $-\frac{2}{3}$ (B) $-\frac{1}{3}$ (C) $\frac{4}{3}$ (D) $\frac{1}{3}$

Open solution

Q163. JEE Main 2016 (Online) 10th April Morning Slot

The solution of the differential equation

$\frac{dy}{dx} + \frac{y}{2}\sec x = \frac{\tan x}{2y}$,

where $0 \le x < \frac{\pi}{2}$, and y (0) = 1, is given by :

(A) $y = 1 - \frac{x}{\sec x + \tan x}$ (B) $y^2 = 1 + \frac{x}{\sec x + \tan x}$

(C) $y^2 = 1 - \frac{x}{\sec x + \tan x}$ (D) $y = 1 + \frac{x}{\sec x + \tan x}$

Open solution

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Q164. JEE Main 2016 (Online) 9th April Morning Slot

If f(x) is a differentiable function in the interval $(0, \infty)$ such that f (1) = 1 and

$\lim_{t \to x} \frac{t^2 f(x) - x^2 f(t)}{t - x} = 1$, for each x > 0, then $f\left(\frac{3}{2}\right)$ equal to :

(A) $\frac{13}{6}$ (B) $\frac{23}{18}$ (C) $\frac{25}{9}$ (D) $\frac{31}{18}$

Open solution

Q165. JEE Main 2016 (Offline)

If a curve $y = f(x)$ passes through the point $(1, -1)$ and satisfies the differential equation, $y(1 + xy)dx = x\, dy$, then $f\left(-\frac{1}{2}\right)$ is equal to :

(A) $\frac{2}{5}$ (B) $\frac{4}{5}$ (C) $-\frac{2}{5}$ (D) $-\frac{4}{5}$

Open solution

Q166. JEE Main 2015 (Online) 11th April Morning Slot

The solution of the differential equation

$ydx - (x + 2y^2)\, dy = 0$ is $x = f(y)$.

If $f(-1) = 1$, then $f(1)$ is equal to :

(A) 4 (B) 3 (C) 2 (D) 1

Open solution

Q167. JEE Main 2015 (Online) 10th April Morning Slot

If y(x) is the solution of the differential equation

$(x + 2)\frac{dy}{dx} = x^2 + 4x - 9$, $x \neq -2$ and $y(0) = 0$,

then $y(-4)$ is equal to :

(A) 0 (B) 1 (C) -1 (D) 2

Open solution

Q168. JEE Main 2015 (Offline)

Let $y(x)$ be the solution of the differential equation

$(x \log x)\frac{dy}{dx} + y = 2x \log x, (x \geq 1)$. Then $y(e)$ is equal to :

(A) 2 (B) $2e$ (C) e (D) 0

Open solution

Q169. JEE Main 2014 (Online) 19th April Morning Slot

If $\frac{dy}{dx}$ + y tan x = sin 2x and y(0) = 1, then $y(\pi)$ is equal to :

(A) 1 (B) -1 (C) -5 (D) 5

Open solution

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Q170. JEE Main 2014 (Online) 12th April Morning Slot

The general solution of the differential equation, sin 2x $\left(\frac{dy}{dx} - \sqrt{\tan x}\right)$ − y = 0, is :

(A) y$\sqrt{\tan x}$ = x + c

(B) y$\sqrt{\cot x}$ = tan x + c

(C) y$\sqrt{\tan x}$ = cot x + c

(D) y$\sqrt{\cot x}$ = x + c

Open solution

Q171. JEE Main 2014 (Online) 11th April Morning Slot

If the general solution of the differential equation y' = $\frac{y}{x}$ + $\Phi\left(\frac{x}{y}\right)$, for some function Φ, is given by y ln $|cx|$ = x, where c is an arbitrary constant, then Φ (2) is equal to :

(A) 4 (B) $\frac{1}{4}$ (C) − 4 (D) $-\frac{1}{4}$

Open solution

Q172. JEE Main 2014 (Offline)

Let the population of rabbits surviving at time t be governed by the differential equation $\frac{dp(t)}{dt} = \frac{1}{2}p(t) - 200$. If $p(0) = 100$, then $p(t)$ equals:

(A) $600 - 500\,e^{t/2}$ (B) $400 - 300\,e^{-t/2}$ (C) $400 - 300\,e^{t/2}$ (D) $300 - 200\,e^{-t/2}$

Open solution

Q173. JEE Main 2013 (Online) 25th April Morning Slot

The equation of the curve passing through the origin and satisfying the differential equation $(1 + x^2)\frac{dy}{dx} + 2xy = 4x^2$ is

(A) $(1 + x^2)y = x^3$ (B) $3(1 + x^2)y = 2x^3$ (C) $(1 + x^2)y = 3x^3$ (D) $3(1 + x^2)y = 4x^3$

Open solution

Q174. JEE Main 2013 (Online) 23th April Morning Slot

If a curve passes through the point $(2, \frac{7}{2})$ and has slope $\left(1 - \frac{1}{x^2}\right)$ at any point (x, y) on it, then the ordinate of the point on the curve whose abscissa is -2 is :

(A) $-\frac{3}{2}$ (B) $\frac{3}{2}$ (C) $\frac{5}{2}$ (D) $-\frac{5}{2}$

Open solution

Q175. JEE Main 2013 (Online) 22th April Morning Slot

Consider the differential equation :

$\frac{dy}{dx} = \frac{y^3}{2(xy^2 - x^2)}$

Statement-1: The substitution $z = y^2$ transforms the above equation into a first order homogeneous differential equation.

Statement-2: The solution of this differential equation is $y^2e^{-y^2/x} = C$.

(A) Both statements are false.

(B) Statement-1 is true and statement-2 is false.

(C) Statement-1 is false and statement-2 is true.

(D) Both statements are true.

Open solution

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Q176. JEE Main 2013 (Offline)

At present, a firm is manufacturing 2000 items. It is estimated that the rate of change of production P w.r.t. additional number of workers x is given by $\frac{dp}{dx} = 100 - 12\sqrt{x}$. If the firm employs 25 more workers, then the new level of production of items is

(A) 2500 (B) 3000 (C) 3500 (D) 4500

Open solution

Q177. AIEEE 2012 (Online) 26th May

The integrating factor of the differential equation $\left(x^2 - 1\right)\frac{dy}{dx} + 2xy = x$ is

(A) $\frac{1}{x^2-1}$ (B) $x^2 - 1$ (C) $\frac{x^2-1}{x}$ (D) $\frac{x}{x^2-1}$

Open solution

Q178. AIEEE 2012 (Online) 19th May

The general solution of the differential equation $\frac{dy}{dx} + \frac{2}{x}y = x^2$ is

(A) $y = cx^{-3} - \frac{x^2}{4}$ (B) $y = cx^3 - \frac{x^2}{4}$ (C) $y = cx^2 + \frac{x^3}{5}$ (D) $y = cx^{-2} + \frac{x^3}{5}$

Open solution

Q179. AIEEE 2012 (Online) 12th May

Statement 1: The degrees of the differential equations $\frac{dy}{dx} + y^2 = x$ and $\frac{d^2y}{dx^2} + y = \sin x$ are equal.
Statement 2: The degree of a differential equation, when it is a polynomial equation in derivatives, is the highest positive integral power of the highest order derivative involved in the differential equation, otherwise degree is not defined.

(A) Statement 1 is true, Statement 2 is true; Statement 2 is not a correct explanation of Statement 1.

(B) Statement 1 is false, Statement 2 is true.

(C) Statement 1 is true, Statement 2 is false.

(D) Statement 1 is true, Statement 2 is true; Statement 2 is a correct explanation of Statement 1.

Open solution

Q180. AIEEE 2012 (Online) 7th May

Let $y(x)$ be a solution of $\frac{2+\sin x}{1+y}\frac{dy}{dx} = \cos x$. If $y(0) = 2$, then $y\left(\frac{\pi}{2}\right)$ equals

(A) $\frac{5}{2}$ (B) 2 (C) $\frac{7}{2}$ (D) 3

Open solution

Q181. AIEEE 2012

The population $p\,(t)$ at time t of a certain mouse species satisfies the differential equation $\frac{dp(t)}{dt} = 0.5\,p\,(t) - 450$. If $p(0) = 850$, then the time at which the population becomes zero is :

(A) $2ln\ 18$ (B) $ln\ 9$ (C) $\frac{1}{2}ln\ 18$ (D) $ln\ 18$

Open solution

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Q182. AIEEE 2011

Let I be the purchase value of an equipment and $V(t)$ be the value after it has been used for t years. The value $V(t)$ depreciates at a rate given by differential equation $\frac{dV(t)}{dt} = -k(T-t)$, where $k > 0$ is a constant and T is the total life in years of the equipment. Then the scrap value $V(T)$ of the equipment is

(A) $I - \frac{kT^2}{2}$

(B) $I - \frac{k(T-t)^2}{2}$

(C) e^{-kT}

(D) $T^2 - \frac{1}{k}$

Open solution

Q183. AIEEE 2011

If $\frac{dy}{dx} = y + 3 > 0$ and $y(0) = 2$, then $y(\ln 2)$ is equal to :

(A) 5 (B) 13 (C) -2 (D) 7

Open solution

Q184. AIEEE 2010

Solution of the differential equation

$\cos x\, dy = y(\sin x - y)dx,\ 0 < x < \frac{\pi}{2}$ is :

(A) $y \sec x = \tan x + c$

(B) $y \tan x = \sec x + c$

(C) $\tan x = (\sec x + c)y$

(D) $\sec x = (\tan x + c)y$

Open solution

Q185. AIEEE 2008

The solution of the differential equation

$\frac{dy}{dx} = \frac{x+y}{x}$ satisfying the condition $y(1) = 1$ is :

(A) $y = \ln x + x$

(B) $y = x \ln x + x^2$

(C) $y = xe^{(x-1)}$

(D) $y = x\ \ln x + x$

Open solution

Q186. AIEEE 2006

The differential equation whose solution is $Ax^2 + By^2 = 1$ where A and B are arbitrary constants is of

(A) second order and second degree

(B) first order and second degree

(C) first order and first degree

(D) second order and first degree

Open solution

Q187. AIEEE 2005

The differential equation representing the family of curves $y^2 = 2c\left(x + \sqrt{c}\right)$, where $c > 0$, is a parameter, is of order and degree as follows:

(A) order 1, degree 2 (B) order 1, degree 1 (C) order 1, degree 3 (D) order 2, degree 2

Open solution

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Q188. AIEEE 2005

If $x\frac{dy}{dx} = y(\log y - \log x + 1)$, then the solution of the equation is :

(A) $y\log\left(\frac{x}{y}\right) = cx$

(B) $x\log\left(\frac{y}{x}\right) = cy$

(C) $\log\left(\frac{y}{x}\right) = cx$

(D) $\log\left(\frac{x}{y}\right) = cy$

Open solution

Q189. AIEEE 2004

Solution of the differential equation $ydx + (x + x^2y)dy = 0$ is

(A) $log\, y = Cx$

(B) $-\frac{1}{xy} + \log y = C$

(C) $\frac{1}{xy} + \log y = C$

(D) $-\frac{1}{xy} = C$

Open solution

Q190. AIEEE 2003

The solution of the differential equation

$(1 + y^2) + \left(x - e^{\tan^{-1}y}\right)\frac{dy}{dx} = 0$, is :

(A) $xe^{2\tan^{-1}y} = e^{\tan^{-1}y} + k$

(B) $(x - 2) = ke^{2\tan^{-1}y}$

(C) $2xe^{\tan^{-1}y} = e^{2\tan^{-1}y} + k$

(D) $xe^{\tan^{-1}y} = \tan^{-1}y + k$

Open solution

Q191. AIEEE 2003

The degree and order of the differential equation of the family of all parabolas whose axis is x-axis, are respectively.

(A) $2, 3$

(B) $2, 1$

(C) $1, 2$

(D) $3, 2.$

Open solution

Q192. AIEEE 2002

The order and degree of the differential equation $\left(1 + 3\frac{dy}{dx}\right)^{2/3} = 4\frac{d^3y}{dx^3}$ are

(A) $\left(1, \frac{2}{3}\right)$

(B) $(3, 1)$

(C) $(3, 3)$

(D) $(1, 2)$

Open solution

Q193. AIEEE 2002

The solution of the equation $\frac{d^2y}{dx^2} = e^{-2x}$

(A) $\frac{e^{-2x}}{4}$

(B) $\frac{e^{-2x}}{4} + cx + d$

(C) $\frac{1}{4}e^{-2x} + cx^2 + d$

(D) $\frac{1}{4}e^{-4x} + cx + d$

Open solution

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Numerical 50 questions

Q194. JEE Main 2026 (Online) 6th April Morning Shift

Let $y = y(x)$ be the solution of the differential equation $\left(x^2 - x\sqrt{x^2-1}\right)dy + \left(y\left(x - \sqrt{x^2-1}\right) - x\right)dx = 0, x \geq 1$. If $y(1) = 1$, then the greatest integer less than $y(\sqrt{5})$ is ____ .

Open solution

Q195. JEE Main 2026 (Online) 5th April Evening Shift

Let $y = y(x)$ be the solution of the differential equation $(\tan x)^{1/2}\, dy = \left(\sec^3 x - (\tan x)^{3/2} y\right)dx, 0 < x < \frac{\pi}{2}, y\left(\frac{\pi}{4}\right) = \frac{6\sqrt{2}}{5}$. If $y\left(\frac{\pi}{3}\right) = \frac{4}{5}\alpha$, then α^4 equals ____ .

Open solution

Q196. JEE Main 2026 (Online) 5th April Morning Shift

Let $y = y(x)$ be the solution of the differential equation $x \sin\left(\frac{y}{x}\right)dy = \left(y\sin\left(\frac{y}{x}\right) - x\right)dx, y(1) = \frac{\pi}{2}$ and let $\alpha = \cos\left(\frac{y(e^{12})}{e^{12}}\right)$. Then the number of integral value of p, for which the equation $x^2 + y^2 - 2px + 2py + \alpha + 2 = 0$ represents a circle of radius $r \leq 6$, is ____ .

Open solution

Q197. JEE Main 2026 (Online) 4th April Evening Shift

Let f be a twice differentiable function such that

$f(x) = \int_0^x \tan(t-x)dt - \int_0^x f(t)\tan t dt, x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

Then $f''\left(\frac{\pi}{6}\right) + 12f'\left(-\frac{\pi}{6}\right) + f\left(\frac{\pi}{6}\right)$ is equal to _____.

Open solution

Q198. JEE Main 2026 (Online) 23rd January Evening Shift

If the solution curve $y = f(x)$ of the differential equation

$\left(x^2-4\right)y' - 2xy + 2x\left(4 - x^2\right)^2 = 0, x > 2,$

passes through the point $(3, 15)$, then the local maximum value of f is ____

Open solution

Q199. JEE Main 2026 (Online) 23rd January Morning Shift

Let f be a twice differentiable non-negative function such that $(f(x))^2 = 25 + \int_0^x \left((f(t))^2 + (f'(t))^2\right)\mathrm{dt}$. Then the mean of $f\left(\log_e(1)\right), f\left(\log_e(2)\right), \ldots\ldots, f\left(\log_e(625)\right)$ is equal to ____ .

Open solution

Q200. JEE Main 2025 (Online) 2nd April Evening Shift

Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} + 2y\sec^2 x = 2\sec^2 x + 3\tan x \cdot \sec^2 x$ such that $y(0) = \frac{5}{4}$. Then $12\left(y\left(\frac{\pi}{4}\right) - e^{-2}\right)$ is equal to_____

Open solution

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Q201. JEE Main 2025 (Online) 28th January Evening Shift

If $y = y(x)$ is the solution of the differential equation, $\sqrt{4-x^2}\frac{dy}{dx} = \left(\left(\sin^{-1}\left(\frac{x}{2}\right)\right)^2 - y\right)\sin^{-1}\left(\frac{x}{2}\right), -2 \le x \le 2, y(2) = \frac{\pi^2-8}{4}$, then $y^2(0)$ is equal to _____.

Open solution

Q202. JEE Main 2025 (Online) 24th January Evening Shift

Let $y = y(x)$ be the solution of the differential equation

$2\cos x\frac{dy}{dx} = \sin 2x - 4y\sin x, x \in \left(0, \frac{\pi}{2}\right)$. If $y\left(\frac{\pi}{3}\right) = 0$, then $y'\left(\frac{\pi}{4}\right) + y\left(\frac{\pi}{4}\right)$ is equal to _____.

Open solution

Q203. JEE Main 2025 (Online) 24th January Morning Shift

Let f be a differentiable function such that $2(x+2)^2 f(x) - 3(x+2)^2 = 10\int_0^x (t+2)f(t)dt, x \ge 0$. Then $f(2)$ is equal to _____ .

Open solution

Q204. JEE Main 2025 (Online) 22nd January Evening Shift

Let $y = f(x)$ be the solution of the differential equation $\frac{dy}{dx} + \frac{xy}{x^2-1} = \frac{x^6+4x}{\sqrt{1-x^2}}, -1 < x < 1$ such that $f(0) = 0$. If $6\int_{-1/2}^{1/2} f(x)dx = 2\pi - \alpha$ then α^2 is equal to _____ .

Open solution

Q205. JEE Main 2024 (Online) 9th April Evening Shift

For a differentiable function $f : \mathbb{R} \to \mathbb{R}$, suppose $f'(x) = 3f(x) + \alpha$, where $\alpha \in \mathbb{R}, f(0) = 1$ and $\lim\limits_{x\to-\infty} f(x) = 7$. Then $9f(-\log_e 3)$ is equal to _____.

Open solution

Q206. JEE Main 2024 (Online) 8th April Evening Shift

Let $\alpha|x| = |y|e^{xy-\beta}, \alpha, \beta \in \mathbf{N}$ be the solution of the differential equation $x\,dy - y\,dx + xy(x\,dy + y\,dx) = 0, y(1) = 2$. Then $\alpha + \beta$ is equal to _____

Open solution

Q207. JEE Main 2024 (Online) 6th April Evening Shift

If the solution $y(x)$ of the given differential equation $(e^y + 1)\cos x\,dx + e^y \sin x\,dy = 0$ passes through the point $\left(\frac{\pi}{2}, 0\right)$, then the value of $e^{y\left(\frac{\pi}{6}\right)}$ is equal to _____.

Open solution

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Q208. JEE Main 2024 (Online) 5th April Evening Shift

Let $y = y(x)$ be the solution of the differential equation

$\frac{dy}{dx} + \frac{2x}{(1+x^2)^2}y = xe^{\frac{1}{(1+x^2)}}; y(0) = 0.$

Then the area enclosed by the curve $f(x) = y(x)e^{-\frac{1}{(1+x^2)}}$ and the line $y - x = 4$ is _____.

Open solution

Q209. JEE Main 2024 (Online) 4th April Evening Shift

Let $y = y(x)$ be the solution of the differential equation $(x + y + 2)^2 dx = dy, y(0) = -2$. Let the maximum and minimum values of the function $y = y(x)$ in $\left[0, \frac{\pi}{3}\right]$ be α and β, respectively. If $(3\alpha + \pi)^2 + \beta^2 = \gamma + \delta\sqrt{3}, \gamma, \delta \in \mathbb{Z}$, then $\gamma + \delta$ equals _____.

Open solution

Q210. JEE Main 2024 (Online) 4th April Morning Shift

Let the solution $y = y(x)$ of the differential equation $\frac{dy}{dx} - y = 1 + 4\sin x$ satisfy $y(\pi) = 1$. Then $y\left(\frac{\pi}{2}\right) + 10$ is equal to _____.

Open solution

Q211. JEE Main 2024 (Online) 1st February Evening Shift

If $\frac{dx}{dy} = \frac{1+x-y^2}{y}, x(1) = 1$, then $5x(2)$ is equal to _____.

Open solution

Q212. JEE Main 2024 (Online) 1st February Morning Shift

If $x = x(t)$ is the solution of the differential equation $(t+1)dx = \left(2x + (t+1)^4\right)dt, x(0) = 2$, then, $x(1)$ equals _____.

Open solution

Q213. JEE Main 2024 (Online) 31st January Evening Shift

Let $y = y(x)$ be the solution of the differential equation

$\sec^2 x dx + \left(e^{2y}\tan^2 x + \tan x\right)dy = 0, 0 < x < \frac{\pi}{2}, y(\pi/4) = 0.$

If $y(\pi/6) = \alpha$, then $e^{8\alpha}$ is equal to _____.

Open solution

Q214. JEE Main 2024 (Online) 30th January Evening Shift

Let $Y = Y(X)$ be a curve lying in the first quadrant such that the area enclosed by the line $Y - y = Y'(x)(X - x)$ and the co-ordinate axes, where (x, y) is any point on the curve, is always $\frac{-y^2}{2Y'(x)} + 1, Y'(x) \neq 0$. If $Y(1) = 1$, then $12Y(2)$ equals _____.

Open solution

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Q215. JEE Main 2024 (Online) 30th January Morning Shift

Let $y = y(x)$ be the solution of the differential equation $(1-x^2)\mathrm{d}y = \left[xy + (x^3+2)\sqrt{3(1-x^2)}\right]\mathrm{d}x, -1 < x < 1, y(0) = 0$. If $y\left(\frac{1}{2}\right) = \frac{\mathrm{m}}{\mathrm{n}}$, m and n are co-prime numbers, then $\mathrm{m}+\mathrm{n}$ is equal to _____.

Open solution

Q216. JEE Main 2024 (Online) 29th January Morning Shift

If the solution curve $y = y(x)$ of the differential equation $\left(1+y^2\right)(1+\log_{\mathrm{e}} x)dx + xdy = 0, x > 0$ passes through the point $(1,1)$ and $y(e) = \frac{\alpha - \tan\left(\frac{3}{2}\right)}{\beta + \tan\left(\frac{3}{2}\right)}$, then $\alpha + 2\beta$ is _____.

Open solution

Q217. JEE Main 2024 (Online) 27th January Evening Shift

If the solution curve, of the differential equation $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x+y-2}{x-y}$ passing through the point $(2,1)$ is $\tan^{-1}\left(\frac{y-1}{x-1}\right) - \frac{1}{\beta}\log_{\mathrm{e}}\left(\alpha + \left(\frac{y-1}{x-1}\right)^2\right) = \log_{\mathrm{e}}|x-1|$, then $5\beta+\alpha$ is equal to _____.

Open solution

Q218. JEE Main 2024 (Online) 27th January Morning Shift

If the solution of the differential equation

$(2x+3y-2)\mathrm{d}x + (4x+6y-7)\mathrm{d}y = 0, y(0) = 3$, is

$\alpha x + \beta y + 3\log_e|2x+3y-\gamma| = 6$, then $\alpha + 2\beta + 3\gamma$ is equal to _____.

Open solution

Q219. JEE Main 2023 (Online) 13th April Evening Shift

If $y = y(x)$ is the solution of the differential equation

$\frac{dy}{dx} + \frac{4x}{(x^2-1)}y = \frac{x+2}{(x^2-1)^{\frac{5}{2}}}, x > 1$ such that

$y(2) = \frac{2}{9}\log_e(2+\sqrt{3})$ and $y(\sqrt{2}) = \alpha\log_e(\sqrt{\alpha}+\beta) + \beta - \sqrt{\gamma}, \alpha, \beta, \gamma \in \mathbb{N}$, then $\alpha\beta\gamma$ is equal to :

Open solution

Q220. JEE Main 2023 (Online) 10th April Evening Shift

Let the tangent at any point P on a curve passing through the points (1, 1) and $\left(\frac{1}{10}, 100\right)$, intersect positive x-axis and y-axis at the points A and B respectively. If $\mathrm{PA} : \mathrm{PB} = 1 : k$ and $y = y(x)$ is the solution of the differential equation $e^{\frac{dy}{dx}} = kx + \frac{k}{2}, y(0) = k$, then $4y(1) - 6\log_{\mathrm{e}} 3$ is equal to _____.

Open solution

Q221. JEE Main 2023 (Online) 8th April Evening Shift

Let the solution curve $x = x(y), 0 < y < \frac{\pi}{2}$, of the differential equation $\left(\log_e(\cos y)\right)^2 \cos y\,\mathrm{d}x - (1 + 3x\log_e(\cos y))\sin y dy = 0$ satisfy $x\left(\frac{\pi}{3}\right) = \frac{1}{2\log_e 2}$. If $x\left(\frac{\pi}{6}\right) = \frac{1}{\log_e m - \log_e n}$, where m and n are coprime, then mn is equal to _____.

Open solution

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Q222. JEE Main 2023 (Online) 8th April Morning Shift

If the solution curve of the differential equation $(y - 2\log_e x)dx + (x\log_e x^2)dy = 0, x > 1$ passes through the points $(e, \frac{4}{3})$ and (e^4, α), then α is equal to _____.

Open solution

Q223. JEE Main 2023 (Online) 6th April Morning Shift

Let $y = y(x)$ be a solution of the differential equation $(x\cos x)dy + (xy\sin x + y\cos x - 1)dx = 0, 0 < x < \frac{\pi}{2}$. If $\frac{\pi}{3}y\left(\frac{\pi}{3}\right) = \sqrt{3}$, then $\left|\frac{\pi}{6}y''\left(\frac{\pi}{6}\right) + 2y'\left(\frac{\pi}{6}\right)\right|$ is equal to _____.

Open solution

Q224. JEE Main 2022 (Online) 27th July Morning Shift

Let $y = y(x)$ be the solution curve of the differential equation $\sin\left(2x^2\right)\log_e\left(\tan x^2\right)dy + \left(4xy - 4\sqrt{2}x\sin\left(x^2 - \frac{\pi}{4}\right)\right)dx = 0, 0 < x < \sqrt{\frac{\pi}{2}}$, which passes through the point $\left(\sqrt{\frac{\pi}{6}}, 1\right)$. Then $\left|y\left(\sqrt{\frac{\pi}{3}}\right)\right|$ is equal to _____.

Open solution

Q225. JEE Main 2022 (Online) 26th July Evening Shift

Suppose $y = y(x)$ be the solution curve to the differential equation $\frac{dy}{dx} - y = 2 - e^{-x}$ such that $\lim\limits_{x\to\infty} y(x)$ is finite. If a and b are respectively the x - and y-intercepts of the tangent to the curve at $x = 0$, then the value of $a - 4b$ is equal to _____.

Open solution

Q226. JEE Main 2022 (Online) 25th July Evening Shift

Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} = \frac{4y^3 + 2yx^2}{3xy^2 + x^3}, y(1) = 1$.

If for some $n \in \mathbb{N}, y(2) \in [n-1, n)$, then n is equal to _____.

Open solution

Q227. JEE Main 2022 (Online) 30th June Morning Shift

A hostel has 100 students. On a certain day (consider it day zero) it was found that two students are infected with some virus. Assume that the rate at which the virus spreads is directly proportional to the product of the number of infected students and the number of non-infected students. If the number of infected students on 4th day is 30, then number of infected students on 8th day will be _____.

Open solution

Q228. JEE Main 2022 (Online) 29th June Evening Shift

Let y = y(x), x > 1, be the solution of the differential equation $(x-1)\frac{dy}{dx} + 2xy = \frac{1}{x-1}$, with $y(2) = \frac{1+e^4}{2e^4}$. If $y(3) = \frac{e^\alpha + 1}{\beta e^\alpha}$, then the value of $\alpha + \beta$ is equal to _____.

Open solution

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Q229. JEE Main 2022 (Online) 29th June Morning Shift

Let y = y(x) be the solution of the differential equation $\frac{dy}{dx} + \frac{\sqrt{2}y}{2\cos^4 x - \cos 2x} = xe^{\tan^{-1}(\sqrt{2}\cot 2x)}$, $0 < x < \frac{\pi}{2}$ with $y\left(\frac{\pi}{4}\right) = \frac{\pi^2}{32}$. If $y\left(\frac{\pi}{3}\right) = \frac{\pi^2}{18}e^{-\tan^{-1}(\alpha)}$, then the value of $3\alpha^2$ is equal to _____.

Open solution

Q230. JEE Main 2022 (Online) 27th June Evening Shift

Let $y = y(x)$ be the solution of the differential equation $(1-x^2)dy = \left(xy + (x^3+2)\sqrt{1-x^2}\right)dx$, $-1 < x < 1$, and $y(0) = 0$. If $\int_{\frac{-1}{2}}^{\frac{1}{2}} \sqrt{1-x^2}y(x)dx = k$, then k $^{-1}$ is equal to _____.

Open solution

Q231. JEE Main 2022 (Online) 26th June Morning Shift

Let the solution curve y = y(x) of the differential equation

$(4+x^2)dy - 2x(x^2+3y+4)dx = 0$ pass through the origin. Then y(2) is equal to _____.

Open solution

Q232. JEE Main 2022 (Online) 26th June Morning Shift

Let $S = (0, 2\pi) - \left\{\frac{\pi}{2}, \frac{3\pi}{4}, \frac{3\pi}{2}, \frac{7\pi}{4}\right\}$. Let $y = y(x)$, x ∈ S, be the solution curve of the differential equation $\frac{dy}{dx} = \frac{1}{1+\sin 2x}$, $y\left(\frac{\pi}{4}\right) = \frac{1}{2}$. If the sum of abscissas of all the points of intersection of the curve y = y(x) with the curve $y = \sqrt{2}\sin x$ is $\frac{k\pi}{12}$, then k is equal to _____.

Open solution

Q233. JEE Main 2021 (Online) 27th July Evening Shift

Let y = y(x) be the solution of the differential equation dy = $e^{\alpha x + y}$ dx; $\alpha \in$ N. If y($\log_e 2$) = $\log_e 2$ and y(0) = $\log_e\left(\frac{1}{2}\right)$, then the value of α is equal to _____.

Open solution

Q234. JEE Main 2021 (Online) 27th July Morning Shift

If $y = y(x), y \in \left[0, \frac{\pi}{2}\right)$ is the solution of the differential equation $\sec y\frac{dy}{dx} - \sin(x+y) - \sin(x-y) = 0$, with y(0) = 0, then $5y'\left(\frac{\pi}{2}\right)$ is equal to _____.

Open solution

Q235. JEE Main 2021 (Online) 25th July Morning Shift

Let y = y(x) be solution of the following differential equation $e^y\frac{dy}{dx} - 2e^y\sin x + \sin x\cos^2 x = 0, y\left(\frac{\pi}{2}\right) = 0$ If $y(0) = \log_e(\alpha + \beta e^{-2})$, then $4(\alpha+\beta)$ is equal to _____.

Open solution

Q236. JEE Main 2021 (Online) 22th July Evening Shift

Let y = y(x) be the solution of the differential equation $\left((x+2)e^{\left(\frac{y+1}{x+2}\right)} + (y+1)\right)dx = (x+2)dy$, y(1) = 1. If the domain of y = y(x) is an open interval (α, β), then | α + β| is equal to _____.

Open solution

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Q237. JEE Main 2021 (Online) 20th July Evening Shift

Let a curve y = y(x) be given by the solution of the differential equation $\cos\left(\frac{1}{2}\cos^{-1}(e^{-x})\right)dx = \sqrt{e^{2x}-1}dy$. If it intersects y-axis at y = -1, and the intersection point of the curve with x-axis is $(\alpha, 0)$, then e^{α} is equal to _____.

Open solution

Q238. JEE Main 2021 (Online) 18th March Evening Shift

Let y = y(x) be the solution of the differential equation

$xdy - ydx = \sqrt{(x^2-y^2)}dx$, $x \geq 1$, with y(1) = 0. If the area bounded by the line x = 1, x = e^{π}, y = 0 and y = y(x) is $\alpha e^{2\pi} + \beta$, then the value of $10(\alpha + \beta)$ is equal to _____.

Open solution

Q239. JEE Main 2021 (Online) 16th March Morning Shift

Let the curve y = y(x) be the solution of the differential equation, $\frac{dy}{dx}$ = 2(x + 1). If the numerical value of area bounded by the curve y = y(x) and x-axis is $\frac{4\sqrt{8}}{3}$, then the value of y(1) is equal to _____.

Open solution

Q240. JEE Main 2021 (Online) 26th February Morning Shift

The difference between degree and order of a differential equation that represents the family of curves given by $y^2 = a\left(x + \frac{\sqrt{a}}{2}\right)$, a > 0 is _____.

Open solution

Q241. JEE Main 2021 (Online) 26th February Morning Shift

If y = y(x) is the solution of the equation

$e^{\sin y}\cos y\frac{dy}{dx} + e^{\sin y}\cos x = \cos x$, y(0) = 0; then

$1 + y\left(\frac{\pi}{6}\right) + \frac{\sqrt{3}}{2}y\left(\frac{\pi}{3}\right) + \frac{1}{\sqrt{2}}y\left(\frac{\pi}{4}\right)$ is equal to _____.

Open solution

Q242. JEE Main 2021 (Online) 25th February Evening Shift

If the curve, y = y(x) represented by the solution of the differential equation $(2xy^2 - y)dx + xdy = 0$, passes through the intersection of the lines, $2x - 3y = 1$ and $3x + 2y = 8$, then |y(1)| is equal to _____.

Open solution

Q243. JEE Main 2020 (Online) 9th January Morning Slot

If for $x \geq 0$, y = y(x) is the solution of the differential equation $(x + 1)dy = ((x + 1)^2 + y - 3)dx$, y(2) = 0, then y(3) is equal to _____.

Open solution

Out of syllabus 23 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

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Q244. JEE Main 2023 (Online) 10th April Morning Shift · MCQ (Single Correct Answer)

The slope of tangent at any point (x, y) on a curve $y = y(x)$ is $\frac{x^2+y^2}{2xy}, x > 0$. If $y(2) = 0$, then a value of $y(8)$ is :

(A) $-4\sqrt{2}$ (B) $2\sqrt{3}$ (C) $4\sqrt{3}$ (D) $-2\sqrt{3}$

Open solution

Q245. JEE Main 2023 (Online) 31st January Morning Shift · MCQ (Single Correct Answer)

Let a differentiable function f satisfy $f(x) + \int_3^x \frac{f(t)}{t}dt = \sqrt{x+1}, x \geq 3$. Then $12f(8)$ is equal to :

(A) 19 (B) 34 (C) 17 (D) 1

Open solution

Q246. JEE Main 2022 (Online) 28th July Evening Shift · MCQ (Single Correct Answer)

The differential equation of the family of circles passing through the points $(0, 2)$ and $(0, -2)$ is :

(A) $2xy\frac{dy}{dx} + (x^2 - y^2 + 4) = 0$ (B) $2xy\frac{dy}{dx} + (x^2 + y^2 - 4) = 0$

(C) $2xy\frac{dy}{dx} + (y^2 - x^2 + 4) = 0$ (D) $2xy\frac{dy}{dx} - (x^2 - y^2 + 4) = 0$

Open solution

Q247. JEE Main 2022 (Online) 25th July Evening Shift · MCQ (Single Correct Answer)

Let a smooth curve $y = f(x)$ be such that the slope of the tangent at any point (x, y) on it is directly proportional to $\left(\frac{-y}{x}\right)$. If the curve passes through the points $(1, 2)$ and $(8, 1)$, then $\left|y\left(\frac{1}{8}\right)\right|$ is equal to

(A) $2\log_e 2$ (B) 4 (C) 1 (D) $4\log_e 2$

Open solution

Q248. JEE Main 2022 (Online) 30th June Morning Shift · MCQ (Single Correct Answer)

Let $\frac{dy}{dx} = \frac{ax-by+a}{bx+cy+a}$, $a, b, c \in R$, represents a circle with center (α, β). Then, $\alpha + 2\beta$ is equal to :

(A) -1 (B) 0 (C) 1 (D) 2

Open solution

Q249. JEE Main 2022 (Online) 27th June Morning Shift · MCQ (Single Correct Answer)

Let $\frac{dy}{dx} = \frac{ax-by+a}{bx+cy+a}$, where a, b, c are constants, represent a circle passing through the point (2, 5). Then the shortest distance of the point (11, 6) from this circle is :

(A) 10 (B) 8 (C) 7 (D) 5

Open solution

Q250. JEE Main 2022 (Online) 24th June Evening Shift · MCQ (Single Correct Answer)

The slope of normal at any point (x, y), x > 0, y > 0 on the curve y = y(x) is given by $\frac{x^2}{xy-x^2y^2-1}$. If the curve passes through the point (1, 1), then e . y(e) is equal to

(A) $\frac{1-\tan(1)}{1+\tan(1)}$ (B) tan(1)

(C) 1 (D) $\frac{1+\tan(1)}{1-\tan(1)}$

Open solution

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Q251. JEE Main 2021 (Online) 27th August Evening Shift · MCQ (Single Correct Answer)

A differential equation representing the family of parabolas with axis parallel to y-axis and whose length of latus rectum is the distance of the point (2, −3) from the line 3x + 4y = 5, is given by :

(A) $10\frac{d^2y}{dx^2} = 11$ (B) $11\frac{d^2x}{dy^2} = 10$ (C) $10\frac{d^2x}{dy^2} = 11$ (D) $11\frac{d^2y}{dx^2} = 10$

Open solution

Q252. JEE Main 2021 (Online) 18th March Morning Shift · MCQ (Single Correct Answer)

The differential equation satisfied by the system of parabolas

$y^2 = 4a(x + a)$ is :

(A) $y\left(\frac{dy}{dx}\right)^2 - 2x\left(\frac{dy}{dx}\right) - y = 0$

(B) $y\left(\frac{dy}{dx}\right)^2 - 2x\left(\frac{dy}{dx}\right) + y = 0$

(C) $y\left(\frac{dy}{dx}\right)^2 + 2x\left(\frac{dy}{dx}\right) - y = 0$

(D) $y\left(\frac{dy}{dx}\right) + 2x\left(\frac{dy}{dx}\right) - y = 0$

Open solution

Q253. JEE Main 2021 (Online) 24th February Evening Shift · MCQ (Single Correct Answer)

If a curve y = f(x) passes through the point (1, 2) and satisfies $x\frac{dy}{dx} + y = bx^4$, then for what value of b, $\int_1^2 f(x)dx = \frac{62}{5}$?

(A) $\frac{31}{5}$ (B) 10 (C) 5 (D) $\frac{62}{5}$

Open solution

Q254. JEE Main 2020 (Online) 6th September Evening Slot · MCQ (Single Correct Answer)

If $y = \left(\frac{2}{\pi}x - 1\right)cosec\, x$ is the solution of the differential equation,

$\frac{dy}{dx} + p(x)y = \frac{2}{\pi}cosec\, x$,

$0 < x < \frac{\pi}{2}$, then the function p(x) is equal to :

(A) cot x (B) sec x (C) tan x (D) cosec x

Open solution

Q255. JEE Main 2020 (Online) 8th January Evening Slot · MCQ (Single Correct Answer)

The differential equation of the family of curves, $x^2 = 4b(y + b)$, $b \in R$, is :

(A) $x(y')^2 = x - 2yy'$ (B) $x(y')^2 = 2yy' - x$

(C) $xy'' = y'$ (D) $x(y')^2 = x + 2yy'$

Open solution

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Q256. JEE Main 2019 (Online) 8th April Evening Slot · MCQ (Single Correct Answer)

Given that the slope of the tangent to a curve y = y(x) at any point (x, y) is $\frac{2y}{x^2}$. If the curve passes through the centre of the circle $x^2 + y^2 - 2x - 2y = 0$, then its equation is :

(A) $x \log_e|y| = 2(x - 1)$

(B) $x^2 \log_e|y| = -2(x - 1)$

(C) $x \log_e|y| = x - 1$

(D) $x \log_e|y| = -2(x - 1)$

Open solution

Q257. JEE Main 2018 (Online) 16th April Morning Slot · MCQ (Single Correct Answer)

The differential equation representing the family of ellipse having foci either on the x-axis or on the y-axis, center at the origin and passing through the point (0, 3) is :

(A) $xy\,y'' + x\,(y')^2 - y\,y' = 0$

(B) $x + y\,y'' = 0$

(C) $xy\,y' + y^2 - 9 = 0$

(D) $xy\,y' - y^2 + 9 = 0$

Open solution

Q258. JEE Main 2014 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

If the differential equation representing the family of all circles touching x-axis at the origin is $(x^2 - y^2)\frac{dy}{dx} = g(x)\,y$, then g(x) equals :

(A) $\frac{1}{2}x$

(B) $2x^2$

(C) $2x$

(D) $\frac{1}{2}x^2$

Open solution

Q259. JEE Main 2013 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

Statement-1: The slope of the tangent at any point P on a parabola, whose axis is the axis of x and vertex is at the origin, is inversely proportional to the ordinate of the point P.
Statement-2: The system of parabolas $y^2 = 4ax$ satisfies a differential equation of degree 1 and order 1.

(A) Statement-1 is true; Statement-2 is true; Statement-2 is a correct explanation for statement-1.

(B) Statement-1 is true; Statement-2 is true; Statement-2 is not a correct explanation for statement-1.

(C) Statement-1 is true; Statement-2 is false.

(D) Statement-1 is false; Statement-2 is true.

Open solution

Q260. AIEEE 2009 · MCQ (Single Correct Answer)

The differential equation which represents the family of curves $y = c_1 e^{c_2 x}$, where c_1 , and c_2 are arbitrary constants, is

(A) $y'' = y'y$

(B) $yy'' = y'$

(C) $yy'' = (y')^2$

(D) $y' = y^2$

Open solution

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Q261. AIEEE 2008 · MCQ (Single Correct Answer)

The differential equation of the family of circles with fixed radius 5 units and centre on the line $y = 2$ is :

(A) $(x-2)y'^2 = 25-(y-2)^2$

(B) $(y-2)y'^2 = 25-(y-2)^2$

(C) $(y-2)^2y'^2 = 25-(y-2)^2$

(D) $(x-2)^2y'^2 = 25-(y-2)^2$

Open solution

Q262. AIEEE 2007 · MCQ (Single Correct Answer)

The differential equation of all circles passing through the origin and having their centres on the x-axis is :

(A) $y^2 = x^2 + 2xy\frac{dy}{dx}$

(B) $y^2 = x^2 - 2xy\frac{dy}{dx}$

(C) $x^2 = y^2 + xy\frac{dy}{dx}$

(D) $x^2 = y^2 + 3xy\frac{dy}{dx}$

Open solution

Q263. AIEEE 2004 · MCQ (Single Correct Answer)

The differential equation for the family of circle $x^2 + y^2 - 2ay = 0$, where a is an arbitrary constant is :

(A) $\left(x^2+y^2\right)y' = 2xy$

(B) $2\left(x^2+y^2\right)y' = xy$

(C) $\left(x^2-y^2\right)y' = 2xy$

(D) $2\left(x^2-y^2\right)y' = xy$

Open solution

Q264. JEE Main 2022 (Online) 26th July Morning Shift · Numerical

Let a curve $y = y(x)$ pass through the point $(3, 3)$ and the area of the region under this curve, above the x-axis and between the abscissae 3 and $x(> 3)$ be $\left(\frac{y}{x}\right)^3$. If this curve also passes through the point $(\alpha, 6\sqrt{10})$ in the first quadrant, then α is equal to _____.

Open solution

Q265. JEE Main 2021 (Online) 27th August Morning Shift · Numerical

If $y^{1/4} + y^{-1/4} = 2x$, and

$(x^2-1)\frac{d^2y}{dx^2} + \alpha x\frac{dy}{dx} + \beta y = 0$, then $|\alpha - \beta|$ is equal to _____.

Open solution

Q266. JEE Main 2021 (Online) 25th July Evening Shift · Numerical

Let a curve y = f(x) pass through the point $(2, (\log_e 2)^2)$ and have slope $\frac{2y}{x\log_e x}$ for all positive real value of x. Then the value of f(e) is equal to _____.

Open solution

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Answer Key

MCQ (Single Correct Answer)

1. **A**	2. **C**	3. **A**	4. **D**	5. **C**
6. **B**	7. **C**	8. **D**	9. **C**	10. **D**
11. **D**	12. *Scan QR*	13. **B**	14. **A**	15. **B**
16. **B**	17. **B**	18. **C**	19. **C**	20. **A**
21. **A**	22. **B**	23. **D**	24. **A**	25. **A**
26. **D**	27. **B**	28. **C**	29. **D**	30. **C**
31. **A**	32. **D**	33. **A**	34. **C**	35. **D**
36. **B**	37. **D**	38. **C**	39. **A**	40. **B**
41. **D**	42. **C**	43. **D**	44. **D**	45. **B**
46. **D**	47. **A**	48. **C**	49. **C**	50. **B**
51. **D**	52. **B**	53. **A**	54. **B**	55. **B**
56. **A**	57. **C**	58. **A**	59. **A**	60. **A**
61. **D**	62. **A**	63. **A**	64. **D**	65. **D**
66. **C**	67. **B**	68. **D**	69. **B**	70. **A**
71. **A**	72. **D**	73. **C**	74. **D**	75. **A**
76. **A**	77. **B**	78. **B**	79. **D**	80. **B**
81. **A**	82. **A**	83. **B**	84. **A**	85. **B**
86. **A**	87. **C**	88. **A**	89. **D**	90. **B**
91. **A**	92. **A**	93. **B**	94. **D**	95. **D**
96. **C**	97. **B**	98. **B**	99. **C**	100. **A**
101. **D**	102. **A**	103. **B**	104. **B**	105. **D**
106. **A**	107. **C**	108. **C**	109. **D**	110. **B**
111. **A**	112. **A**	113. **D**	114. **C**	115. **A**
116. **A**	117. **B**	118. **D**	119. **A**	120. **C**
121. **B**	122. **B**	123. **D**	124. **A**	125. **D**
126. **B**	127. **C**	128. **C**	129. **B**	130. **A**
131. **A**	132. **C**	133. **D**	134. **D**	135. **B**
136. **D**	137. **D**	138. **C**	139. **D**	140. **C**
141. **A**	142. **A**	143. **B**	144. **B**	145. **D**
146. **C**	147. **B**	148. **C**	149. **D**	150. **D**

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151. **D**	152. **C**	153. **C**	154. **A**	155. **A**
156. **B**	157. **D**	158. **A**	159. **D**	160. **B**
161. **B**	162. **D**	163. **C**	164. **D**	165. **B**
166. **B**	167. **A**	168. **A**	169. **C**	170. **D**
171. **D**	172. **C**	173. **D**	174. **A**	175. **D**
176. **C**	177. **B**	178. **D**	179. **D**	180. **C**
181. **A**	182. **A**	183. **D**	184. **D**	185. **D**
186. **D**	187. **C**	188. **C**	189. **B**	190. **C**
191. **C**	192. **C**	193. **B**		

Numerical

194. **3**	195. **48**	196. **11**	197. **5**	198. **16**
199. **1565**	200. **21**	201. **4**	202. **1**	203. **19**
204. **27**	205. **61**	206. **4**	207. **3**	208. **18**
209. **31**	210. **7**	211. **5**	212. **14**	213. **9**
214. **20**	215. **97**	216. **3**	217. **11**	218. **29**
219. **6**	220. **4**	221. **12**	222. **3**	223. **2**
224. **1**	225. **3**	226. **3**	227. **90**	228. **14**
229. **2**	230. **320**	231. **12**	232. **42**	233. **2**
234. **2**	235. **4**	236. **4**	237. **2**	238. **4**
239. **2**	240. **2**	241. **1**	242. **1**	243. **3**

Out of syllabus

244. **C**	245. **C**	246. **A**	247. **B**	248. **C**
249. **B**	250. **D**	251. **D**	252. **C**	253. **B**
254. **A**	255. **D**	256. **A**	257. **D**	258. **C**
259. **B**	260. **C**	261. **C**	262. **A**	263. **C**
264. **6**	265. **17**	266. **1**		

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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